

LECTURE 2 · IMMUNE MOLECULES

Antibody and Complement

Structure · Diversity · Function · The Three Pathways · Regulation · Disease

PART 1 The Antibody Molecule — Two Ends, Two Jobs

PART 2 Where Diversity Comes From

PART 3 The Five Classes and What They Do

PART 4 Making and Using Antibodies

PART 5 Complement — The Cast and the Three Pathways

PART 6 Complement — Regulation, Function, Disease

PART 7 Master Tables, Self-Test and Glossary

SYNTHESIZED FROM

“Immune molecules: Antibody and Complement” — Lecture 2

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Abbas, Lichtman & Pillai *Cellular and Molecular Immunology* · Basic Immunology · Janeway's Immunobiology · Murphy & Weaver

For the mechanisms the slides state but do not explain

How to Use This Guide

Immunology is hard for a specific reason: it is a very large cast of molecules with confusingly similar names, acting on each other in a fixed order. **C3a and C3b are different molecules. CH2 and CH3 do different jobs. CR1 and CR3 bind different things.** Memorising them as a list fails. This guide is built on a method that makes them derivable instead.

The four questions

Every fact in this session is an answer to one of four questions. Ask them in this order of any molecule you meet, and you will have organised it before you have memorised it:

1. **What is it?** — its structure and where it comes from.
2. **What does it bind?** — its ligand or receptor. Nothing in immunology acts at a distance.
3. **What does it do?** — the consequence of that binding.
4. **What breaks if it is gone?** — the deficiency disease, which is the *proof* of the answer to question 3.

Six devices built on those questions

Device	What it is for
Molecule card	Every major molecule gets the same six rows, in the same order: <i>what it is</i> · <i>where it comes from</i> · <i>what it binds</i> · <i>what it does</i> · <i>how it is controlled</i> · <i>what breaks without it</i> . Cover the lower rows and produce them yourself.
End tags V-END C-END	Every antibody feature is tagged for which end it belongs to — the variable end that recognises , or the constant end that acts . See below.
Pathway tags CP LP AP TP	Every complement component is tagged with the pathway or pathways it belongs to: classical , lectin , alternative , terminal . The commonest exam question is “which component belongs where”, so the answer is carried on the component itself.
Absence proves function	A red box giving the disease that appears when the molecule is missing. In immunology this is not a footnote — it is the cleanest evidence that the stated function is the real one.
Cascade strips	The three complement pathways drawn as step-by-step strips, so you can see what acts on what and where they converge.
Name decoder	Complement notation is systematic, not arbitrary. One table lets you read any symbol you meet — including ones this lecture never mentions.

The organising idea for the antibody half

An antibody is **one molecule doing two unrelated jobs at two different ends**, and almost every fact about it sorts cleanly into one end or the other:

- **V-END** — the **variable** end. Made of VH and VL, contains the CDRs, forms the antigen-binding site, and is the part that differs between one antibody and the next. Its only job is **recognition**.
- **C-END** — the **constant** end. Made of the CH domains, defines the class, and carries every effector function: complement fixation, binding to Fc receptors, transport across placenta and mucosa. Its job is **action**.

Hold that division and the rest follows. *Why does papain give two Fab and one Fc?* Because it cuts above the hinge, separating the two recognition ends from the one action end. *Why does IgG cross the placenta but IgM does not?* Because placental transport is an Fc function and their Fc regions differ. *Why does an idiotype vary but an isotype not?* Because idiotype is a V-end property and isotype is a C-end property.

The organising idea for the complement half

Complement looks like thirty proteins. It is really **one event — the cleavage of C3 — with three different ways of reaching it and one shared consequence afterwards**. Every pathway question reduces to: *what recognises the target, and how does that recognition build a C3 convertase?* After C3, all three pathways are identical.

Suggested study sequence

1. **First pass (≈40 min)**. Parts 1 and 5 only — the structure of the antibody and the cast list of complement. Do not try to learn function yet.
2. **Second pass (≈60 min)**. Parts 2, 3, 4 — diversity, the five classes, production. Then Part 6.
3. **Active recall**. Cover the lower rows of every molecule card and rebuild them. Then draw the three cascade strips from memory on blank paper.
4. **Night before the exam (≈25 min)**. Part 7 — the master tables, then the self-test, then the glossary.

WHAT THE LECTURER SAYS TO TAKE AWAY

The two summary slides name four things, and they are the examinable core:

- The **basic structure** of the antibody.
- The **function** of the antibody.
- “**Review the differences between the three activation pathways of complement.**” — quoted exactly. §7.2 is that review, in one table.
- The **function of complement and its regulation**.

Table of Contents

PART 1 · THE ANTIBODY MOLECULE — TWO ENDS, TWO JOBS

1.1	What an Antibody Is	<u>7</u>
1.2	Heavy and Light Chains	<u>8</u>
1.3	Variable and Constant Regions	<u>9</u>
1.4	CDRs and Framework Regions	<u>10</u>
1.5	The Hinge and the Domain	<u>11</u>
1.6	J Chain and Secretory Piece	<u>13</u>
1.7	Cutting It Up — the Fragments	<u>14</u>

PART 2 · WHERE DIVERSITY COMES FROM

2.1	The Problem of Diversity	<u>17</u>
2.2	V(D)J Recombination	<u>17</u>
2.3	Isotype, Allotype, Idiotype	<u>18</u>

PART 3 · THE FIVE CLASSES AND WHAT THEY DO

3.1	The Five Functions of an Antibody	<u>22</u>
3.2	IgG · IgM · IgA · IgE · IgD	<u>23</u>
3.3	The Class Comparison Table	<u>27</u>

PART 4 · MAKING AND USING ANTIBODIES

4.1	Polyclonal, Monoclonal, Engineered	<u>30</u>
4.2	Antibodies in the Clinic and the Lab	<u>33</u>

PART 5 · COMPLEMENT — THE CAST AND THE THREE PATHWAYS

5.1	What Complement Is, and How to Read Its Names	<u>37</u>
5.2	The Cast List	<u>39</u>
5.3	The Classical Pathway	<u>40</u>
5.4	The Lectin (MBL) Pathway	<u>42</u>
5.5	The Alternative Pathway	<u>43</u>
5.6	The Terminal Pathway and the MAC	<u>45</u>

PART 6 · COMPLEMENT — REGULATION, FUNCTION, DISEASE

6.1	Why Complement Must Be Restrained	<u>48</u>
6.2	The Regulators, One by One	<u>49</u>
6.3	The Six Functions of Complement	<u>53</u>
6.4	Complement and Disease	<u>58</u>

PART 7 · MASTER TABLES, SELF-TEST AND GLOSSARY

7.1	Master Table — The Five Immunoglobulins	<u>63</u>
7.2	Master Table — The Three Pathways Compared	<u>64</u>
7.3	Tell Them Apart — the Look-Alikes	<u>65</u>
7.4	Self-Test — 24 Questions with Answers	<u>66</u>
7.5	Glossary of Key Terms	<u>70</u>

PART ONE

01

The Antibody Molecule

— Two Ends, Two Jobs

One protein, built so that one end recognises and the other end acts. Get that division fixed and every later fact — why papain cuts where it does, why IgG crosses the placenta, why an idiotype varies — falls out of it rather than having to be learned separately.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Draw an IgG monomer and label every region on it.
- Say which chain, which region and which domain carries each property.
- Explain why the hinge is rich in proline and what that buys the molecule.
- Predict the products of papain and pepsin digestion and say what each is used for.

1.1 What an Antibody Is

DEFINITION

An **antibody (Ab)** is a **protein produced by plasma cells in response to immunisation**, which **binds specifically to the particular antigen** that provoked it. An antibody is defined *by* that specific binding — it is not a shape you can recognise in isolation, but a molecule identified by what it sticks to.

DEFINITION

An **immunoglobulin (Ig)** is a **globulin that functions as an antibody, or is similar to one in chemical structure**. Immunoglobulins come in two forms: **secreted Ig (sIg)**, free in plasma and secretions, and **membrane Ig (mIg)**, anchored in the B-cell surface where it serves as the B-cell receptor.

EXAM TRAP — ANTIBODY AND IMMUNOGLOBULIN ARE NOT SYNONYMS

They are used interchangeably in speech, but the definitions differ in an examinable way. **Antibody** is a *functional* term: a molecule that binds a known antigen. **Immunoglobulin** is a *structural* term: any protein with the immunoglobulin fold and chain arrangement. Every antibody is an immunoglobulin; a myeloma protein is an immunoglobulin with no known antigen, so it is *not* usefully called an antibody. The membrane form (mIg) is also an immunoglobulin that is not acting as a free antibody — it is a receptor.

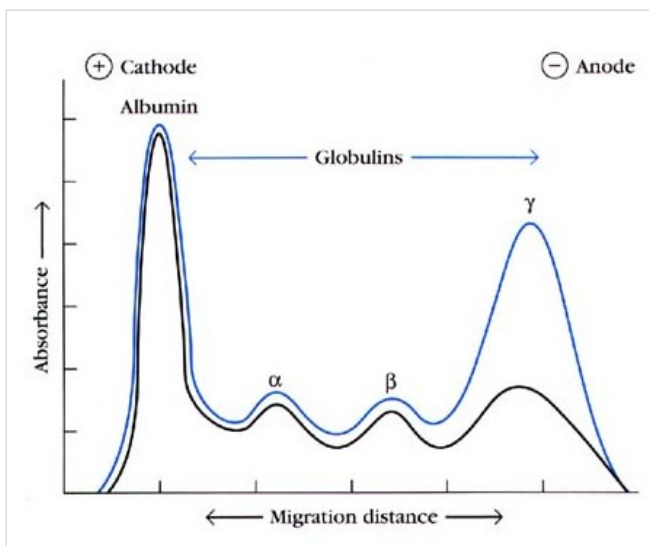


Fig 1.1 Serum protein electrophoresis. Antibodies migrate in the **γ -globulin** fraction — which is where the name immunoglobulin comes from, and why an increase in that peak means increased antibody.

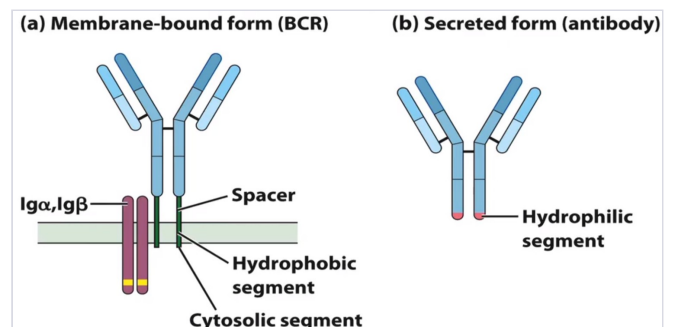


Fig 1.2 The same molecule in two forms. **(a) Membrane form (BCR)**: a hydrophobic transmembrane segment and a short cytosolic tail, associated with $Ig\alpha/Ig\beta$ for signalling. **(b) Secreted form**: the tail is replaced by a hydrophilic terminus and the molecule floats free.

HISTORICAL ANCHOR

Emil von Behring (1854–1917, Germany) took the **first Nobel Prize in Physiology or Medicine, 1901**, for the theory of **antitoxin immunity** — the demonstration that serum from an immunised animal could protect another animal against diphtheria toxin. That serum factor was the antibody, decades before anyone knew its structure. It is worth remembering that **the very first Nobel in medicine was an immunology prize**.

1.2 Heavy and Light Chains

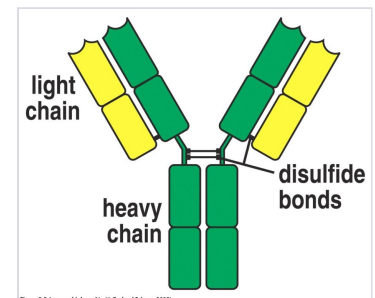


Fig 1.3 The basic unit: **two identical heavy chains and two identical light chains**, held together by **disulfide bonds** — interchain bonds between H and L and between the two H chains, plus intrachain bonds folding each domain.

TABLE 1.1 THE TWO CHAINS

	Heavy chain (H)	Light chain (L)
Mass	50–75 kDa	25 kDa
Length	450–550 amino acids	214 amino acids
Varieties	Five: μ (mu) · γ (gamma) · α (alpha) · ϵ (epsilon) · δ (delta)	Two: κ (kappa) · λ (lambda), with $\lambda 1$ – $\lambda 4$ subtypes
What it determines	The class (isotype) of the whole antibody: $\mu \rightarrow \text{IgM}$, $\gamma \rightarrow \text{IgG}$, $\alpha \rightarrow \text{IgA}$, $\epsilon \rightarrow \text{IgE}$, $\delta \rightarrow \text{IgD}$. Subclasses: IgG1–IgG4 and IgA1–IgA2	Nothing about class. Either κ or λ can pair with any heavy chain, and a given plasma cell uses one or the other, never both

HIGH-YIELD — THE HEAVY CHAIN NAMES THE ANTIBODY

Five Greek letters, five classes, in the same order: **μ γ α ϵ $\delta \rightarrow \text{IgM IgG IgA IgE IgD}$** . The light chain contributes to the binding site but contributes **nothing** to the class or to any effector function. If a question asks what makes IgG different from IgM, the answer is always the heavy chain — never the light chain.

1.3 Variable and Constant Regions

Both chains are divided lengthwise into two regions. The division is the single most important structural fact in this Part:

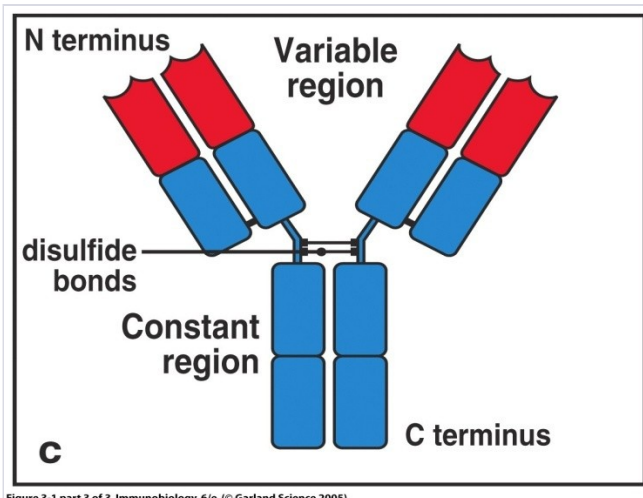


Figure 3-1 part 3 of 3 Immunobiology, 6/e. (© Garland Science 2005)

Fig 1.4 N-terminus at the top, C-terminus at the bottom. The variable region is the amino-terminal end of both chains; the constant region is everything below it. Recognition is at the top; action is at the bottom.

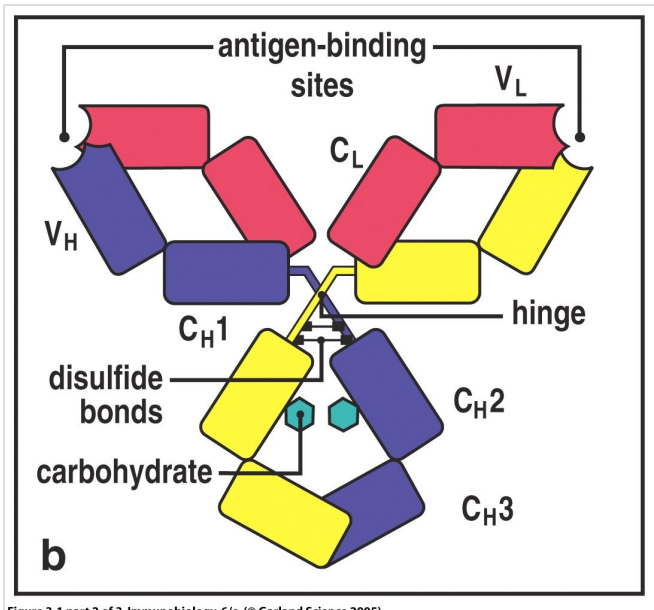


Figure 3-1 part 2 of 3 Immunobiology, 6/e. (© Garland Science 2005)

Fig 1.5 The molecule with every label. Two antigen-binding sites at the top, each made by a V_H and a V_L together; then C_L and C_H1 ; the hinge; then C_H2 and C_H3 with their carbohydrate. Memorise this one figure and Part 1 is largely done.

TABLE 1.2 THE TWO REGIONS, AND WHAT EACH IS FOR

	Variable region V-END	Constant region C-END
Where	Amino-terminal end of both chains	Carboxy-terminal , everything below the V region
Size	VL 110 aa · VH 110 aa — the same length in both chains	CL 110 aa · CH 330–440 aa , made of CH1, CH2, CH3 and in IgM and IgE also CH4
Varies between antibodies?	Yes — this is the only part that does	No — identical in every antibody of the same class
Job	Recognition — VH and VL together form one antigen-binding site	Action — complement fixation, Fc-receptor binding, placental and mucosal transport, half-life
Number per molecule	Two binding sites on a monomer (one per arm)	One Fc

1.4 CDRs and Framework Regions

The variable region is not uniformly variable. Plot the variability along its 110 amino acids and you get a flat baseline interrupted by **three sharp spikes**. Those spikes are where the antigen is actually touched.

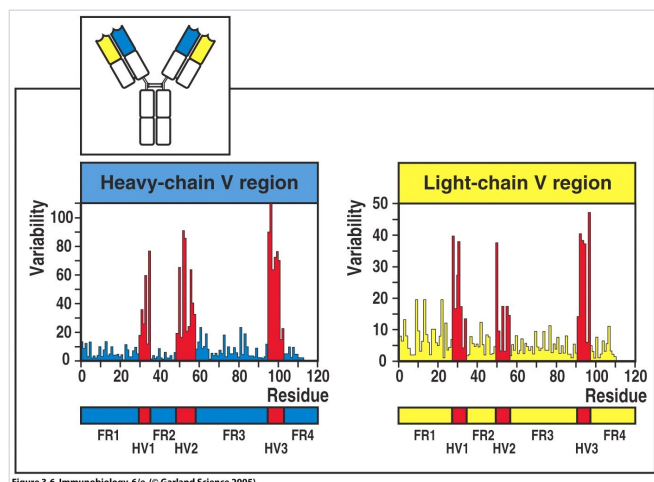


Figure 3-6 Immunobiology, 6/e. (© Garland Science 2005)

Fig 1.6 The Wu-Kabat variability plot, heavy-chain V region (left) and light-chain V region (right). **Three peaks per chain**. The peaks are the **hypervariable regions**; the flat stretches between them are the **framework regions**.

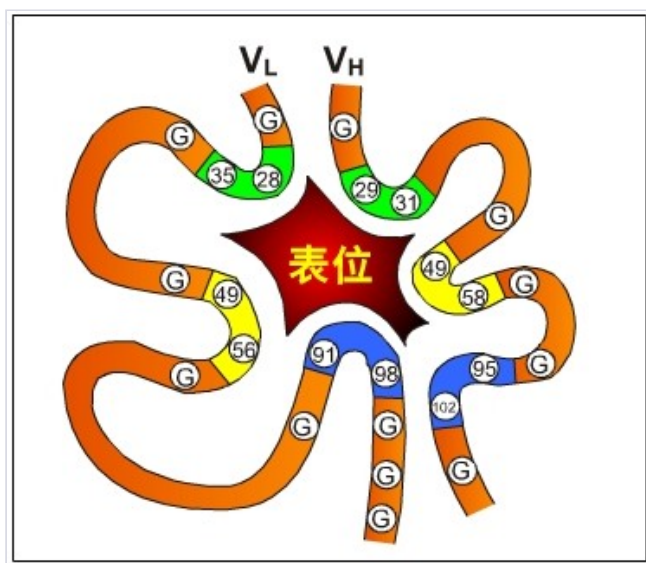


Fig 1.7 Why they are called complementarity-determining. The six loops — three from VH, three from VL — fold together into a surface whose shape is **complementary** to the epitope (表位). The framework holds them in place; the loops do the binding.

TABLE 1.3 INSIDE THE VARIABLE REGION

Term	What it is	What it does
HVR — hypervariable region	The three short stretches of extreme sequence variability in each V region	The sequence-level name for the same thing the structure people call a CDR
CDR — complementarity-determining region	Three per chain, so six per binding site. Present in both H and L chains	Physically contacts the antigen. Their combined shape is what gives an antibody its specificity
FR — framework region	The relatively conserved stretches between the CDRs	Forms the β -sheet scaffold that holds the CDR loops in the right position. It does not touch antigen
Idiotypic	The set of antigenic determinants of the V region — essentially, the CDRs seen as an antigen themselves	Unique to one clone. It is why an antibody can itself be recognised by another antibody — see §2.3

HIGH-YIELD — THE ARITHMETIC OF THE BINDING SITE

3 CDRs per chain $\times$ 2 chains = 6 CDRs per antigen-binding site. 2 binding sites per monomer. The binding site is formed by **VH and VL together** — neither chain alone can bind antigen, which is why a free light chain (Bence Jones protein) has no antibody activity.

1.5 The Hinge and the Domain

REGION · C-END

The hinge region

WHAT IT IS

A short stretch **between CH1 and CH2**.

Rich in proline residues — and in cysteine, which forms the inter-heavy-chain disulfide bonds.

Absent in IgM and IgE, which instead have an extra domain (CH4) in its place.

WHAT IT DOES

Flexibility.

Because: Proline cannot form a regular α -helix or β -sheet — its ring locks the backbone angle and breaks secondary structure. A proline-rich stretch is therefore **necessarily extended and floppy**, and that floppiness lets the two Fab arms swing independently. Two consequences follow. First, one antibody can bridge **two epitopes that are different distances apart**, which no rigid molecule could do. Second, the hinge is the **most exposed part** of the molecule and therefore the site most accessible to proteases — which is exactly why papain and pepsin cut there (§1.7).

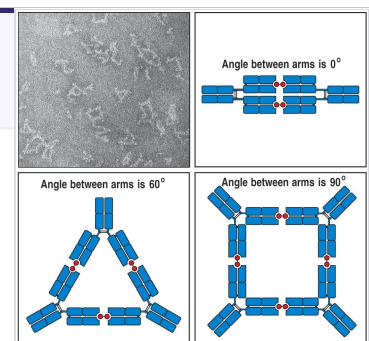


Figure 3-4 Immunobiology, 6/e, © Garland Science 2005

Fig 1.8 What flexibility buys. Top: electron micrographs of antibody–antigen complexes. Middle: the angle between the arms can be **0°, 60° or 180°**. Bottom: because the arms can swing, antibodies and multivalent antigens build triangular, square and larger **lattices** — which is precipitation, agglutination, and the immune complex all at once.

The domain — the repeating unit

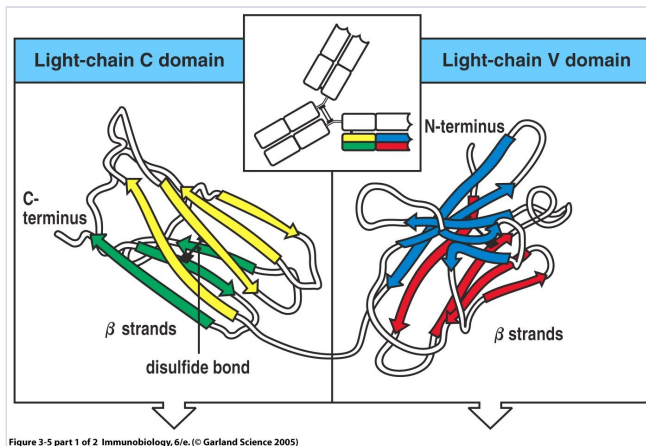


Fig 1.9 A single **domain**: about 110 amino acids folded into two β -sheets facing each other, clamped by an **intrachain disulfide bond**. The whole antibody is built from this unit repeated — VL, CL, VH, CH1, CH2, CH3.

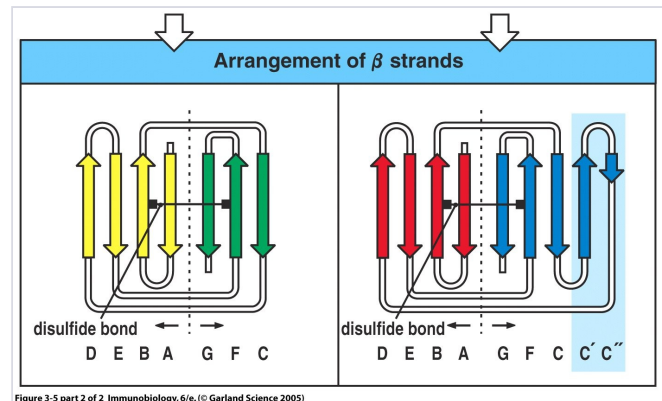


Fig 1.10 The β -strand arrangement differs slightly between the **C domain** (left, seven strands) and the **V domain** (right, nine strands). The two extra strands in the V domain are what carry the CDR loops.

HIGH-YIELD — THE IMMUNOGLOBULIN SUPERFAMILY

The Ig domain is such a useful shape that evolution reused it everywhere. Any protein built from it belongs to the **immunoglobulin superfamily**: the T-cell receptor, MHC class I and II, CD4 and CD8, CD3, Ig α /Ig β , many adhesion molecules, and the poly-Ig receptor that transports IgA. When you meet a new immune molecule described as having “Ig-like domains”, this is what it means — and it means the molecule almost certainly works by **specific binding at the domain tip**.

1.6 J Chain and Secretory Piece

Two classes are not monomers. IgM circulates as a **pentamer** and secreted IgA as a **dimer** — and each needs an extra polypeptide that is not part of the basic four-chain unit.

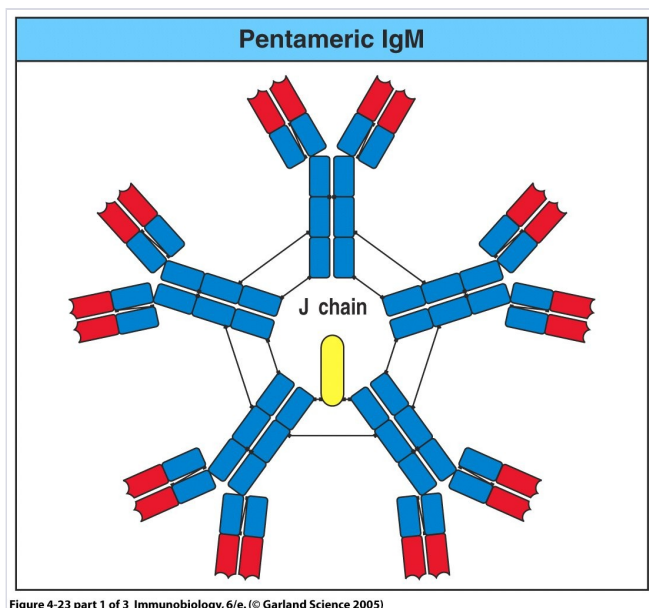


Figure 4-23 part 1 of 3 Immunobiology, 6/e. (© Garland Science 2005)

Fig 1.11 Pentameric IgM. Five monomers arranged in a star, joined at their Fc ends, with a single **J chain** at the centre. Ten binding sites in principle — though steric hindrance means only about five are usable at once against a large antigen.

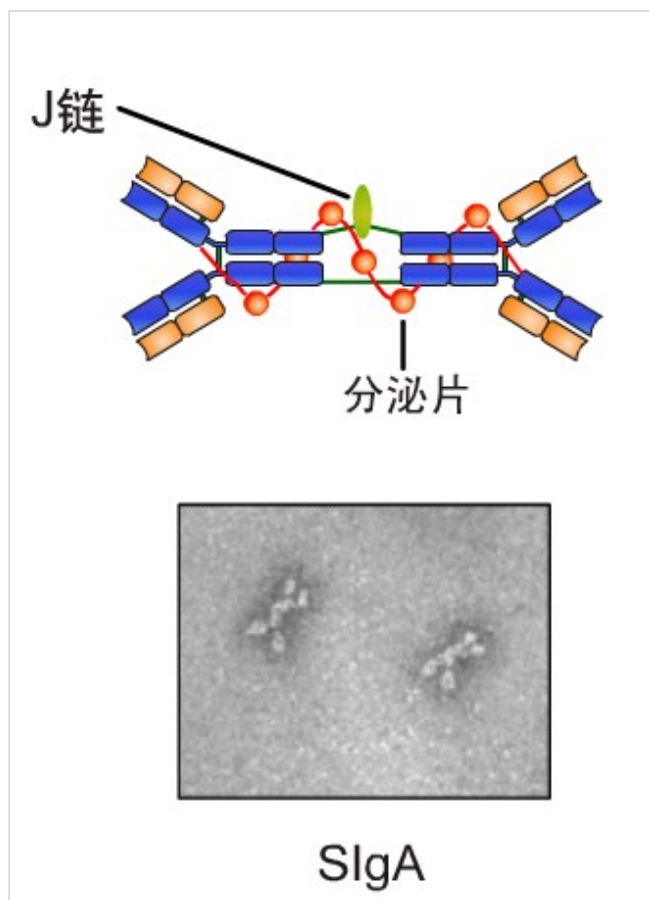


Fig 1.12 Secretory IgA (SIgA). Two monomers joined by a **J chain** (J链), plus the **secretory piece** (分泌片) wrapped around the whole dimer. The electron micrograph below shows the real molecule.

TABLE 1.4 TWO EXTRA POLYPEPTIDES

	J chain (joining chain)	Secretory piece (SP)
What it is	A single small polypeptide chain	A fragment of the poly-Ig receptor — a piece of the epithelial cell's own transport protein
Made by	The plasma cell , along with the antibody	The epithelial cell , not the plasma cell
What it does	Joins the monomer into a polymer — pentameric IgM and dimeric IgA	Carries IgA across the epithelium into the secretion, then stays attached and protects it from proteases in the gut and airway lumen
Found in	IgM and IgA only	Secretory IgA only — serum IgA is a monomer without it

HIGH-YIELD — THE SECRETORY PIECE IS NOT MADE BY THE B CELL

This is asked directly and caught often. The **J chain comes from the plasma cell**; the **secretory piece comes from the epithelial cell** the antibody passes through. It is the leftover extracellular portion of the poly-Ig receptor, cleaved off and carried along. That is why secretory IgA survives in the gut lumen while serum IgA would be digested — it is wearing a piece of the epithelium as armour.

1.7 Cutting It Up — the Fragments

Porter and Edelman took the 1972 Nobel Prize for working out the structure, and the method was simple: **cut the molecule with an enzyme and see what the pieces can still do**. The result is the vocabulary everyone still uses.

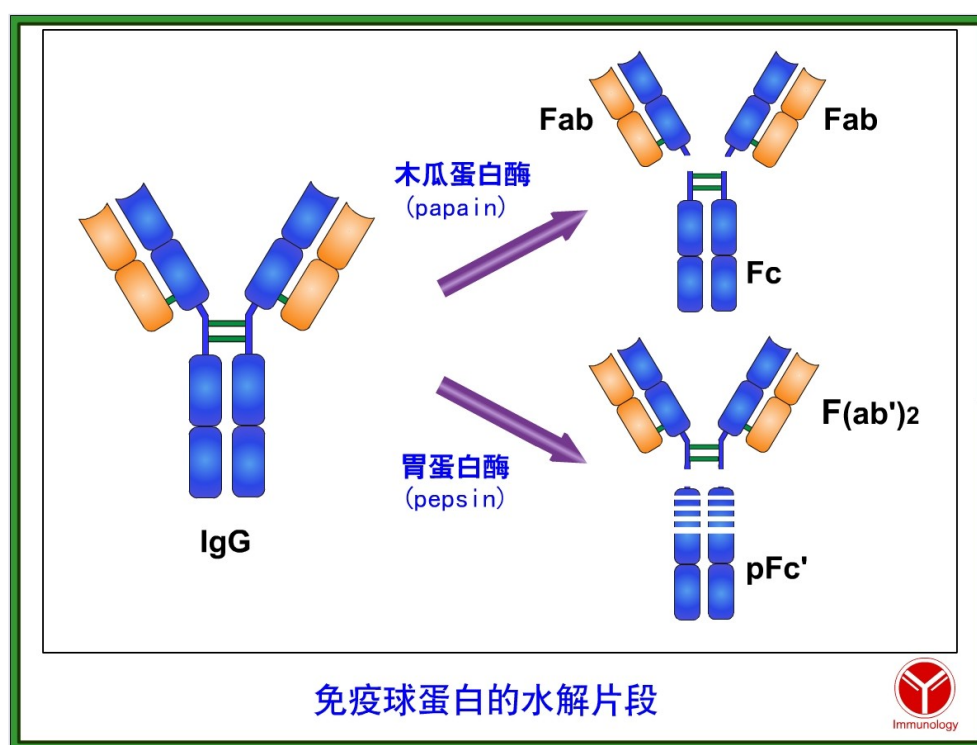


Fig 1.13 Where each enzyme cuts. **Papain** (木瓜蛋白酶) cuts above the inter-heavy-chain disulfide bonds, giving **two separate Fab** and **one Fc**. **Pepsin** (胃蛋白酶) cuts below them, so the two arms stay joined as **F(ab')₂** while the Fc is chopped into useless **pFc'** fragments.

TABLE 1.5 THE FRAGMENTS AND WHAT EACH KEEPS

Fragment	Made by	What it contains	Can it still bind antigen? Can it still act?
Fab V-END	Papain	VL + CL + VH + CH1 — one arm	Binds, monovalently. One binding site, so it attaches but cannot cross-link or precipitate . No effect- or function
Fc C-END	Papain	CH2 + CH3 of both heavy chains, with the carbohydrate	Cannot bind antigen at all. Carries every effector function — it crystallises readily, which is where the “c” comes from

Fragment	Made by	What it contains	Can it still bind antigen? Can it still act?
F(ab')₂ V-END	Pepsin	Both arms, still joined at the hinge	Binds bivalently — it can cross-link, agglutinate and precipitate exactly like whole antibody. But no Fc , so no complement fixation and no Fc-receptor binding
pFc'	Pepsin	Fragments of CH3	Neither. A degradation product with no activity

CLINICAL CORRELATION — WHY F(AB')₂ IS A THERAPEUTIC PRODUCT

Antivenoms and some antitoxins are supplied as **F(ab')₂** rather than whole IgG, and the reason is exactly the table above. You want the binding — bivalent, so it neutralises and clears the toxin efficiently. You do **not** want the Fc, because the Fc of a *horse* antibody injected into a human is a foreign protein that Fc receptors will grab and complement will fix, which is how **serum sickness** and anaphylaxis happen. Cut the Fc off with pepsin and you keep the therapy and lose the reaction.

HIGH-YIELD — REMEMBER IT BY WHERE THE CUT FALLS

Both enzymes cut at the hinge, because the hinge is the exposed, floppy part. The only question is **which side of the inter-chain disulfide bonds**. **Papain cuts above** → the arms come apart → **2 Fab + 1 Fc**. **Pepsin cuts below** → the arms stay together → **1 F(ab')₂ + pFc' debris**. “Papain gives **p**ieces; **p**epsin gives a **p**air.”

PART TWO

02

Where Diversity Comes From

Your genome has about twenty thousand genes. Your antibodies number in the billions of distinct specificities. That is not a rounding error — it is the central problem of immunology, and the answer is that antibody genes are not inherited whole. They are assembled, in each B cell, from pieces.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- State the problem of antibody diversity in numbers.
- List the mechanisms that generate diversity and say which is biggest.
- Distinguish isotype, allotype and idiotype, and say where each lives on the molecule.

2.1 The Problem of Diversity

HIGH-YIELD — THE NUMBER THAT DEFINES THE PROBLEM

The human antibody repertoire contains on the order of 10^{11} **distinct specificities**. The human genome contains about 2×10^4 **genes** in total. There is therefore **no possible way** for one gene to encode one antibody — you would need five million times more genome than you have. Whatever the answer is, it must be **combinatorial**: a small number of parts assembled in a very large number of ways.



Fig 2.1 Susumu Tonegawa, Nobel Prize in Physiology or Medicine **1987**, “for his discovery of the genetic principle for generation of antibody diversity”. He showed that the DNA of a B cell is not the DNA it inherited.

2.2 V(D)J Recombination

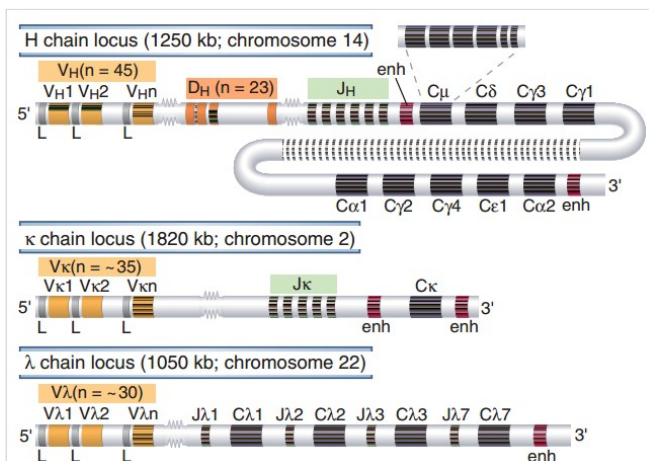


Fig 2.2 The three germline loci. **Heavy chain** on chromosome 14 — V, D, J and C segments. **κ light chain** on chromosome 2 and **λ light chain** on chromosome 22 — V, J and C only, **no D segments**. Every cell in your body is born with these arrays intact.

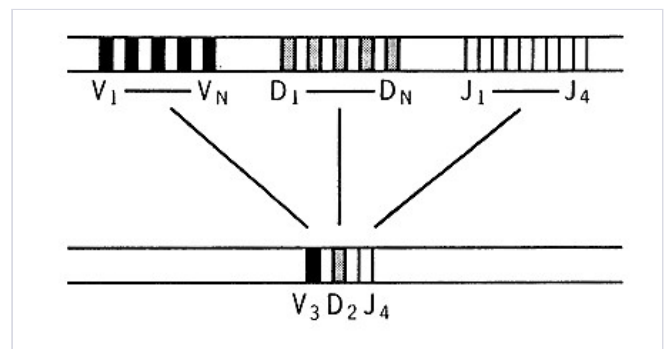


Fig 2.3 The rearrangement. In a developing B cell one **V**, one **D** and one **J** segment are cut out of the array and spliced together at the **DNA** level; everything between them is deleted permanently. The result — V_HD_HJ_H — is a gene that exists in no other cell.

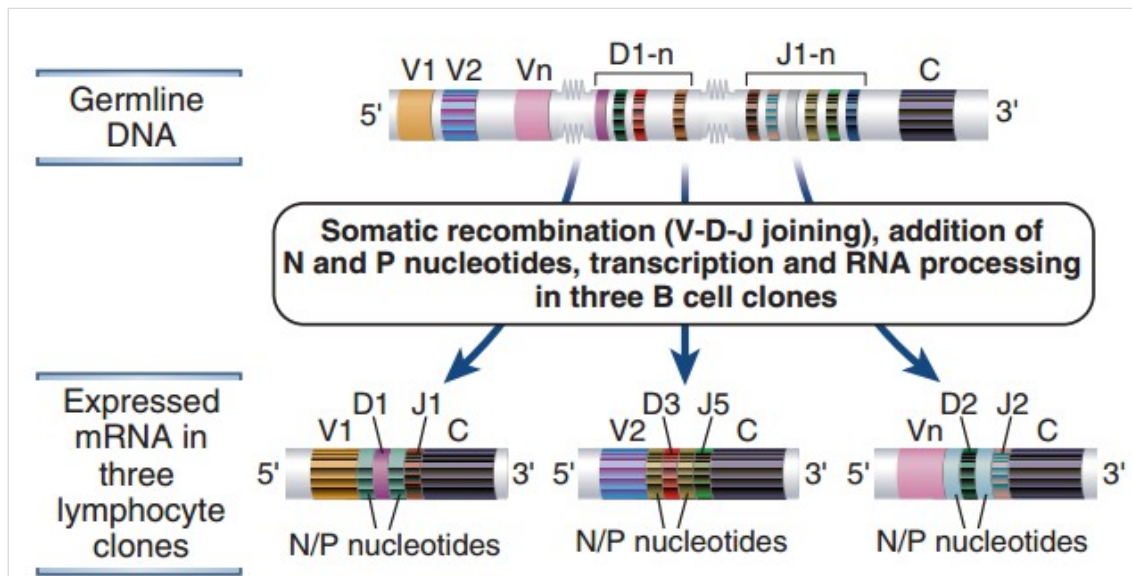


Fig 2.4 From germline DNA to three different antibodies. The same starting array yields different products in three different lymphocyte clones, because **somatic recombination (V-D-J joining)** picks different segments and then **adds or removes nucleotides at the junctions (N/P nucleotides)** before sealing them. Transcription and RNA processing follow.

○ MECHANISM — HOW A SMALL LIBRARY BECOMES A HUGE REPERTOIRE

- 1 **Combinatorial segment choice.** Any V can join any D can join any J.
WHY: With roughly 40 V, 25 D and 6 J segments on the heavy chain, that alone is $40 \times 25 \times 6 \approx 6,000$ heavy-chain V regions from fewer than eighty gene segments.
- 2 **Junctional diversity.** The joins are deliberately sloppy: nucleotides are added (N and P nucleotides) or chewed back at each junction.
WHY: This multiplies the count by orders of magnitude — and, crucially, it falls exactly in **CDR3**, the loop that sits at the centre of the binding site. The single most variable part of an antibody is variable because the enzyme that joins the segments is imprecise on purpose.
- 3 **Heavy–light pairing.** Any rearranged heavy chain can pair with any rearranged light chain.
WHY: Two independent repertoires multiply. Roughly 10^4 heavy $\times 10^3$ light $\approx 10^7$ distinct binding sites before the cell has ever seen an antigen.
- 4 **Somatic hypermutation.** After antigen is encountered, point mutations are introduced into the rearranged V region at a rate a million times the background, and the cells whose mutations improved binding are selected.
WHY: The first three mechanisms build a repertoire **before** exposure — that is why you can respond to a pathogen you have never met. This fourth one refines it **after** exposure, and is why a secondary response has higher affinity than a primary one.

NET RESULT A repertoire of roughly 10^{11} specificities built from a few hundred gene segments — and a B cell whose genome is permanently different from that of every other cell in the body.

2.3 Isotype, Allotype, Idiotype

Three words that sound alike and are constantly confused. They answer three different questions about *where a difference between two antibodies comes from*, and each maps onto a different part of the molecule. The lecture groups them under **heterogeneity of antibody** and separates them by cause: isotype and allotype are **endogenous** — written in the genome — while diversity proper is **exogenous**, induced by the antigen encountered.

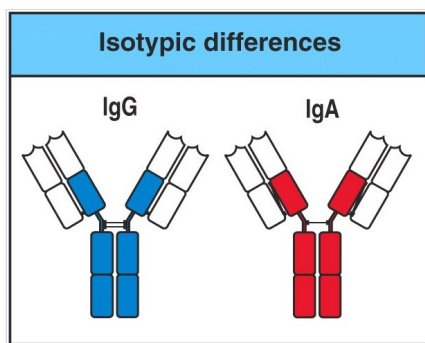


Fig 2.5 Isotypic — IgG versus IgA. The whole **constant region** differs, because they use different heavy-chain genes.

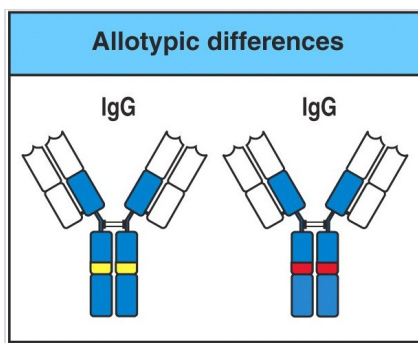


Fig 2.6 Allotypic — IgG versus IgG. Same class; a **small difference within the constant region**, reflecting different alleles in two people.

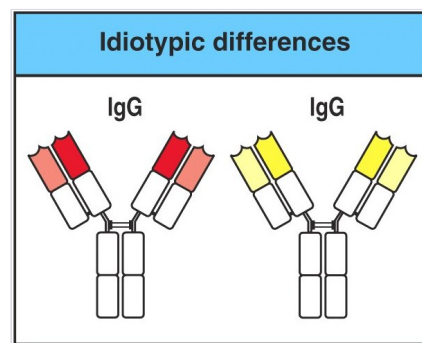


Fig 2.7 Idiotypic — IgG versus IgG, same person. The **variable region** differs, because the two antibodies came from different B-cell clones.

TABLE 2.1 THE THREE KINDS OF VARIATION

	Isotype	Allotype	Idiotype
Where on the molecule	C region C-END	C region C-END	V region , especially the CDRs V-END
Differs between	Classes and subclasses — IgG vs IgA, IgG1 vs IgG3	Individuals of the same species	Clones — even within one individual
Genetic basis	Different genes , all present in every healthy member of the species	Different alleles of the same gene	Different V(D)J rearrangement and somatic mutation
You have	All of them — everyone makes all five classes	Only your own — which is why another person's allotype can be immunogenic to you	Billions of them , one per clone
One-line memory	Iso = same for everyone	Allo = other people	Idio = individual clone

EXAM TRAP — ANTI-IDIOTYPE ANTIBODIES AND WHY THEY EXIST

An idiotype is the **antigenic determinant of the V region** — which is to say, the binding site of one antibody can itself be recognised as an antigen by another antibody. That is not a curiosity: it is the basis of **idiotype network regulation**, and it is why rheumatoid factor (an antibody against the Fc of IgG) and anti-idiotype antibodies are separate phenomena. The examinable point is simply that **an antibody can be an antigen**, and which part gets recognised names the phenomenon: V region → anti-idiotype; C region of another species → anti-isotype; C region of another person → anti-allotype.

PART THREE

03

The Five Classes and What They Do

All five classes have the same V-end and therefore the same capacity to recognise. Everything that distinguishes them is at the C-end — which means every class difference is a difference in *what the antibody does after it has bound*, never in what it can bind.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Name the five functions of an antibody and say which end performs each.
- Give the defining property of each of the five classes.
- Say which classes fix complement, and in which order of strength.
- Explain why IgM dominates a primary response and IgG a secondary one.

3.1 The Five Functions of an Antibody

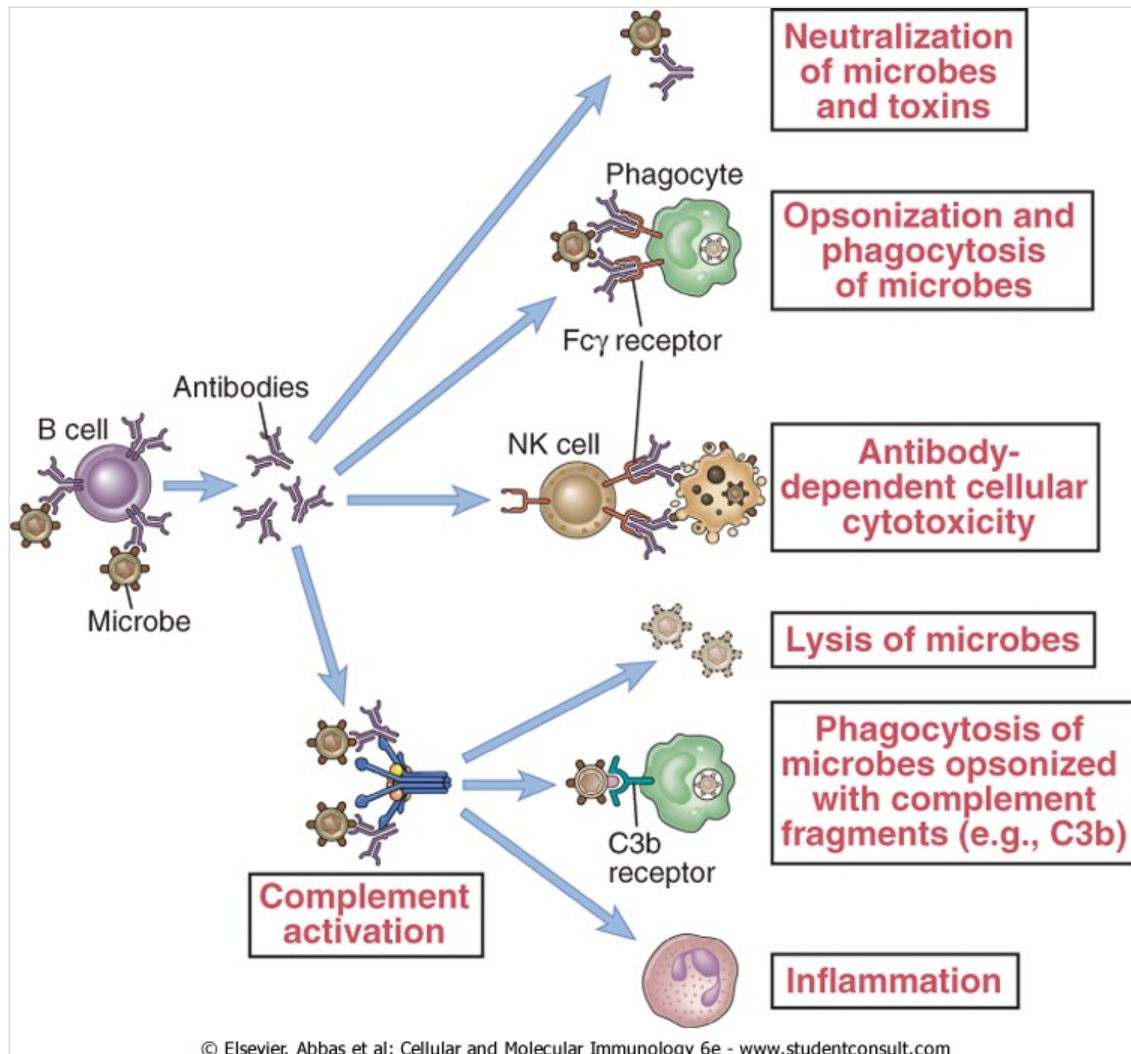


Fig 3.1 Everything an antibody can do, on one diagram. Read it as a fork: the **V-end binds the microbe or toxin**, and then the **Fc decides what happens next** — depending on which cell or molecule picks the Fc up.

TABLE 3.1 THE FIVE FUNCTIONS, AND WHICH END DOES THE WORK

Function	End	How it works
Neutralisation	V-END	Binding alone blocks the toxin's active site or the virus's attachment protein. The only function that needs no Fc at all — which is why an Fab fragment can still neutralise
Opsonisation	C-END	The Fc sticks out from the coated microbe and is grabbed by Fcγ receptors on phagocytes, giving them a handle they otherwise lack
Complement activation	C-END	C1q binds CH2 of IgG or CH3 of IgM — but only once the antibody has bound antigen. See §5.3
ADCC — antibody-dependent cell-mediated cytotoxicity	C-END	An NK cell binds the Fc of antibody already coating a target cell through CD16 (FcγRIII) and kills it. Recognition is done by the antibody; killing by the NK cell

Function	End	How it works
Transport — placenta and mucosa	C-END	FcRn carries IgG across the placenta; the poly-Ig receptor carries IgA across epithelium. Both are Fc-receptor jobs, which is why only certain classes are transported

HIGH-YIELD — THE ANTIBODY DOES NOT KILL ANYTHING

With the single exception of neutralisation, an antibody has **no intrinsic effector activity**. It kills nothing, digests nothing and lyses nothing. What it does is **mark a target and recruit something that can** — complement, a phagocyte, an NK cell. This is why every effector function is an Fc function, and why the class of antibody made determines the kind of defence mounted. Choosing an isotype is choosing a weapon.

3.2 IgG · IgM · IgA · IgE · IgD

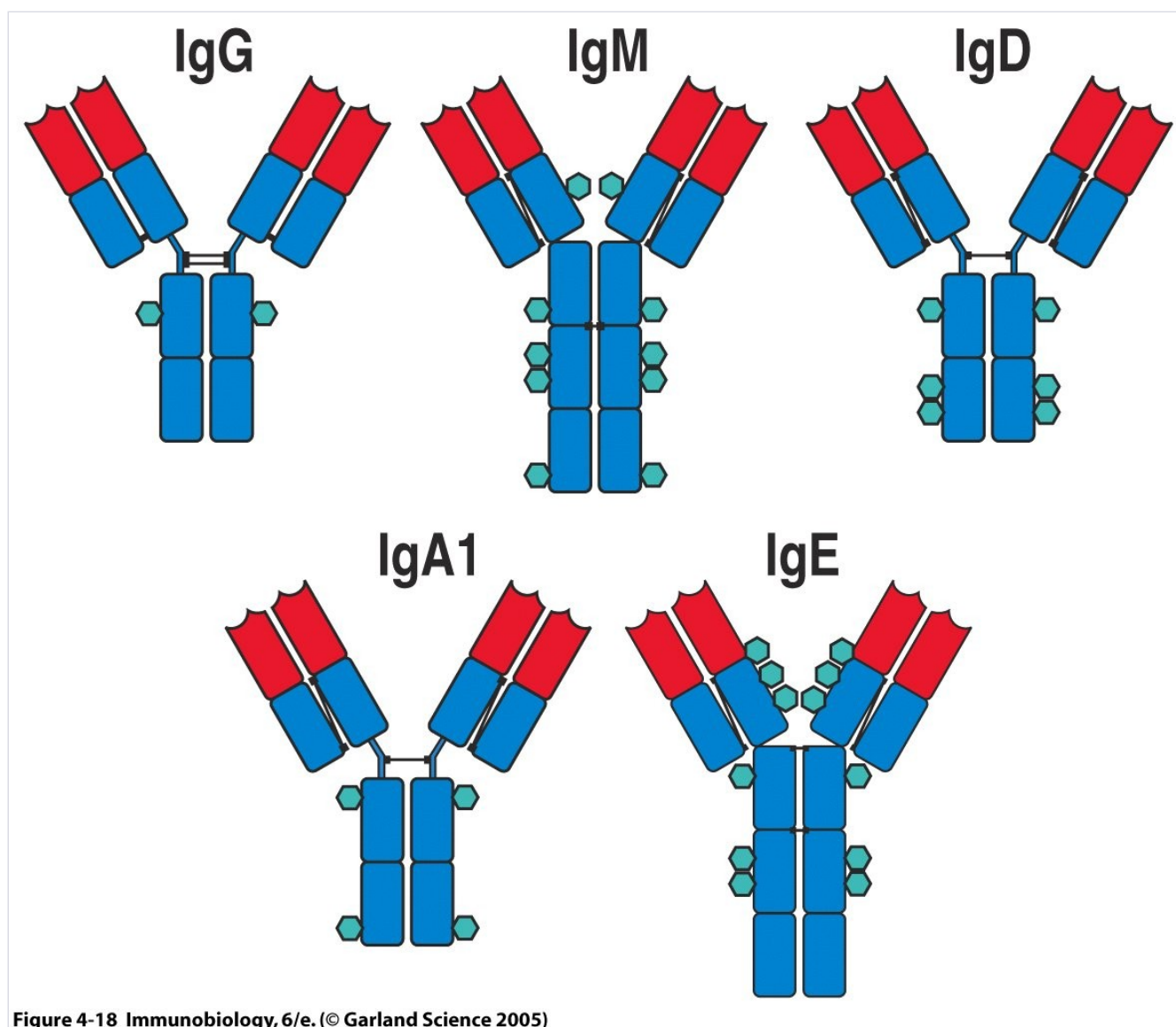


Figure 4-18 Immunobiology, 6/e. (© Garland Science 2005)

Fig 3.2 The five classes drawn to the same scale. Note the structural facts you can read straight off the picture: **IgM is a pentamer**, **IgA1 can be a dimer**, and **IgM and IgE have four CH domains where the others have three plus a hinge**.

CLASS 1 OF 5

IgG — the workhorse

WHAT IT IS

Monomer, γ heavy chain, four subclasses **IgG1–IgG4**. The **most abundant** immunoglobulin in serum (~75–80% of total Ig) and the **longest-lived** (half-life $\approx$ 21 days, far longer than the others).

WHAT BINDS IT

C1q, at **CH2** — IgG1, IgG2 and IgG3 fix complement; **IgG4 does not**.

Fc γ receptors on phagocytes and NK cells.

FcRn on placental syncytiotrophoblast — and on endothelium, which is what gives IgG its long half-life by recycling it out of lysosomes.

WHAT IT DOES

All five functions. **The only class that crosses the placenta** — which is the whole of neonatal humoral immunity for the first months of life.

Because: A newborn cannot make useful antibody for weeks. Maternal IgG transferred across the placenta in the third trimester covers that gap, and its 21-day half-life is exactly what makes the cover last. The physiological trough at about 3–6 months of age, as maternal IgG decays faster than the infant makes its own, is why that is the age at which congenital antibody deficiencies first declare themselves.

REMEMBER IT AS

The **secondary-response** antibody: most abundant, longest-lived, crosses the placenta, does everything.

CLASS 2 OF 5

IgM — the first responder

WHAT IT IS

Pentamer in serum, joined by a **J chain**; a **monomer** when it sits on the B-cell surface as the BCR. μ heavy chain with **four CH domains and no hinge**. The **largest** immunoglobulin — the old name is *macroglobulin*.

WHAT BINDS IT

C1q, at **CH3**. **A single IgM molecule is enough** to start the classical pathway — where IgG needs two side by side.

WHAT IT DOES

The most powerful complement activator of all the classes, and the first antibody made in a primary response.

*Because: Both facts come from the same structure. Being a pentamer, IgM presents **five Fc regions already clustered together**, and C1q needs to engage at least two adjacent Fc sites to be activated. IgG in solution is monomeric and its Fc regions are scattered, so two IgG molecules must happen to land close together on the same surface — a much less likely event. That is also why IgM is efficient at agglutination despite low affinity: ten binding sites give enormous **avidity** even when each individual site binds weakly, which is exactly what a naive response needs before affinity maturation has happened.*

REMEMBER IT AS

The **primary-response** antibody: biggest, first, best at complement, ten binding sites, does not cross the placenta.

CLINICAL CORRELATION — WHY IGM IN CORD BLOOD MEANS INFECTION

IgM does not cross the placenta — it is far too large, and there is no transport receptor for it. So any IgM found in a newborn's blood **must have been made by the newborn**. Detecting IgM against a specific organism in cord blood is therefore diagnostic of **intrauterine infection**, and it is the basis of testing for congenital toxoplasmosis, rubella, CMV and syphilis. Maternal IgG in the same sample proves nothing, because it crossed the placenta and simply reflects the mother's history.

CLASS 3 OF 5**IgA — the mucosal guard****WHAT IT IS**

Monomer in serum; dimer in secretions, joined by a J chain and wrapped in the **secretory piece**. Subclasses IgA1 and IgA2. Made in enormous quantity — more IgA is synthesised per day than all other classes combined, but most of it leaves the body immediately.

WHERE IT IS

Mucosal secretions — saliva, tears, sweat, respiratory and gastrointestinal secretions — and in **colostrum and breast milk**.

WHAT IT DOES

Mucosal immunity: it blocks pathogens at the epithelial surface before they ever enter tissue. The lecture notes that IgA **activates complement via the alternative pathway** — not the classical one.

Because: This is the right weapon for the job. A mucosal surface is not a place to start an inflammatory cascade: inflame the gut wall and you damage the barrier you were defending. So IgA is built to neutralise and exclude rather than to inflame — it does not fix complement classically, and it is a poor opsonin. Breast milk works on exactly this principle: the secretory IgA passed to the infant is not absorbed, it stays in the lumen and coats the gut.

REMEMBER IT AS

The **secretions** antibody: dimer, J chain plus secretory piece, guards mucosa, alternative pathway only.

— ABSENCE PROVES FUNCTION — SELECTIVE IGA DEFICIENCY

The **commonest primary immunodeficiency**, affecting roughly 1 in 600 people. Most are **asymptomatic**, because IgM can be transported into secretions and partly compensate. Those who are not asymptomatic get exactly what the function predicts: **recurrent sinopulmonary and gastrointestinal infections** — surface infections at the surfaces IgA guards. There is also a classic hazard: some patients make **anti-IgA antibodies**, and giving them blood products containing IgA can cause **anaphylaxis**. The disease is a readout of the function, and so is its treatment.

CLASS 4 OF 5**IgE — parasites and allergy****WHAT IT IS**

Monomer, ϵ heavy chain with **four CH domains and no hinge**. Present in serum at the **lowest concentration of all five classes** — thousands of times less than IgG.

WHAT BINDS IT

FcεRI on mast cells and basophils — with such high affinity that IgE is bound essentially permanently, sitting on the mast cell surface waiting.

WHAT IT DOES

Mast-cell degranulation — the immediate (type I) hypersensitivity reaction — and **defence against parasites**.

Because: The mechanism explains both. IgE is pre-loaded onto mast cells; when antigen arrives and **cross-links two adjacent IgE molecules**, the receptors cluster and the cell degranulates within seconds, releasing histamine and other mediators. Against a helminth too large to phagocytose, that sudden local release of vasoactive and smooth-muscle-contracting mediators — plus recruited eosinophils — is a genuinely useful expulsion mechanism. Against pollen it is an allergy. Same machinery, different trigger. This also explains the speed: the antibody was already in place, so no synthesis step stands between exposure and reaction.

REMEMBER IT AS

The **allergy and worm antibody**: rarest in serum, pre-bound to mast cells, cross-linking triggers degranulation.

CLASS 5 OF 5**IgD — the naive B cell's receptor****WHAT IT IS**

Monomer, δ heavy chain. Present in serum in trace amounts, with no established serum function.

WHERE IT IS

On the surface of the **mature naive B cell**, co-expressed with membrane IgM as the **antigen receptor**.

WHAT IT DOES

Acts as a **B-cell receptor**. Its appearance on the surface marks the transition from immature to **mature, naive** B cell — a cell that is ready to respond but has not yet met antigen.

Because: This is the one class whose value is as a **marker rather than a weapon**. If a question asks what surface IgD indicates, the answer is the maturation state of the B cell, not an effector function.

REMEMBER IT AS

The **marker antibody**: surface of the mature naive B cell, no known effector function.

3.3 The Class Comparison Table

Antibody Isotope	Isotype-specific effector functions
IgG	Opsonization of antigens for phagocytosis by macrophages and neutrophils Activation of the classical pathway of complement Antibody-dependent cell-mediated cytotoxicity mediated by natural killer cells Neonatal immunity: transfer of maternal antibody across the placenta and gut Feedback inhibition of B cell activation
IgM	Activation of the classical pathway of complement Antigen receptor of naive B lymphocytes*
IgA	Mucosal immunity: secretion of IgA into the lumens of the gastrointestinal and respiratory tracts Activation of complement by the lectin pathway or by the alternative pathway
IgE	Mast cell degranulation (immediate hypersensitivity reactions)
IgD	Antigen receptor of naive B lymphocytes*

Fig 3.3 The lecture's own effector-function table. Note that every entry is a **C-end** property, which is the point of §3.1 made visible.

HIGH-YIELD — FOUR ONE-LINERS THAT ANSWER MOST CLASS QUESTIONS

- **Most abundant in serum: IgG.** Most abundant *produced* per day: IgA — but it leaves the body.
- **Crosses the placenta: IgG only.** Found in secretions and breast milk: IgA.
- **Best complement activator: IgM** (one molecule suffices), then IgG3 > IgG1 > IgG2. **IgG4, IgA, IgE and IgD do not fix complement classically.**
- **First in a primary response: IgM. Dominant in a secondary response: IgG.** A rising IgM titre means recent or current infection; a high IgG with no IgM means past exposure or vaccination.

PART FOUR

04

Making and Using Antibodies

The immune system makes antibodies by the billion and we can borrow the machinery. Three generations of technology — crude serum, one clone in a dish, and a designed molecule — and each solved a specific problem the previous one had.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Distinguish polyclonal from monoclonal antibody and say when each is preferable.
- Explain what a hybridoma is and why both fusion partners are needed.
- Read a monoclonal antibody's name and predict how immunogenic it is.

4.1 Polyclonal, Monoclonal, Engineered

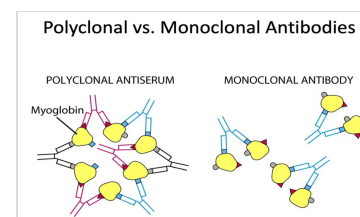


Fig 4.1 The same antigen, two products. **Polyclonal antiserum** contains antibodies from many B-cell clones, binding many different epitopes on the antigen. **Monoclonal antibody** is the product of one clone, binding one epitope — every molecule identical.

TABLE 4.1 THREE WAYS TO GET AN ANTIBODY

	Polyclonal	Monoclonal	Genetically engineered
How	Immunise an animal, bleed it, use the serum	Hybridoma: fuse one immune B cell with a myeloma cell	Clone and modify the V and C genes; express in culture
Specificity	Many epitopes — a mixture	One epitope — chemically uniform	One epitope, and you choose the constant region too
Advantage	Cheap and fast; binds the antigen at many points, so it is robust to small changes in the target	Reproducible for ever — the hybridoma is immortal, so batch 1 and batch 1000 are identical	Low immunogenicity in humans, and effector function can be designed in or out
Limitation	Every batch differs; cross-reactivity; the animal eventually dies	Mouse protein — the patient makes human anti-mouse antibodies (HAMA) against it	Expensive and slow to develop

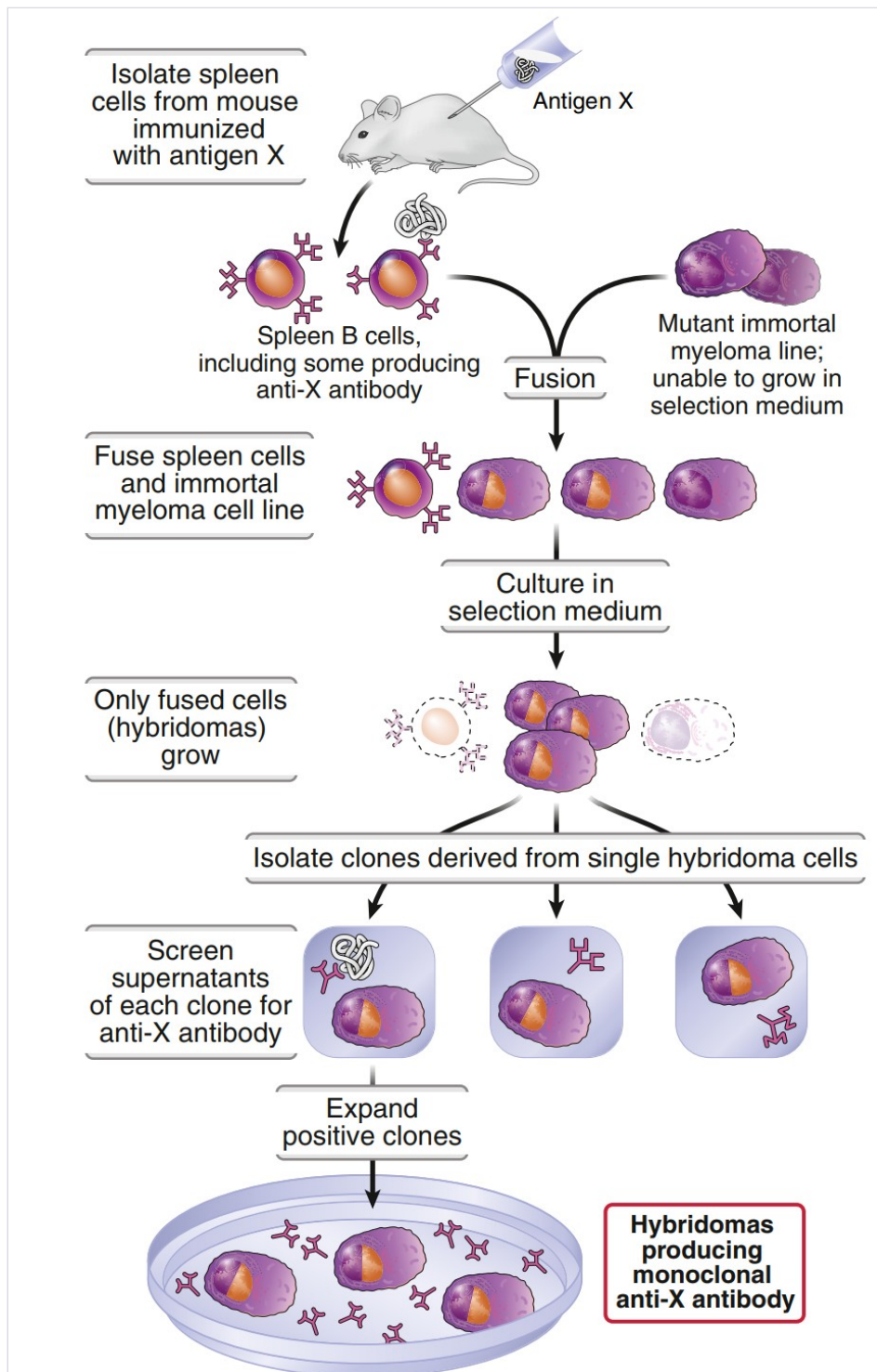


Fig 4.2 Making a hybridoma. Immunise a mouse, take its spleen B cells, **fuse them with myeloma cells**, then grow in **selective medium** in which only the fusions survive. Clone out single cells, screen each clone for the antibody you want, and expand the winner.

○ MECHANISM — WHY FUSION IS NECESSARY, AND WHY SELECTION WORKS

- 1** An immunised mouse's spleen contains **B cells making exactly the antibody you want** — but a B cell in culture **dies within days**.

WHY: You have the specificity but no way to keep it. That is the entire problem the technique solves.
- 2** A **myeloma cell** is a malignant plasma cell: it **grows for ever in culture**, but it makes an antibody of no interest, or none at all.

WHY: You have immortality but no specificity. The two cells have complementary defects.

- 3 **Fuse them.** The hybrid — the **hybridoma** — inherits the B cell's antibody gene and the myeloma's immortality.

WHY: This is the whole idea, and it won the 1984 Nobel Prize precisely because it is so simple in retrospect.

- 4 Grow the mixture in **HAT selective medium**. The myeloma parent is chosen to lack an enzyme of the salvage pathway, so **unfused myeloma cells die; unfused B cells die of old age**; only **hybrids** survive, because the B cell supplies the missing enzyme and the myeloma supplies the immortality.

WHY: The selection is elegant: you do not have to pick out the hybrids, you simply create conditions in which nothing else can live.

- 5 **Clone and screen.** Dilute so that each well contains one cell, let it grow, and test each well's supernatant for the antibody you want.

WHY: This is what makes it monoclonal: every molecule in the final product descends from one B cell, so every molecule is identical.

NET RESULT An immortal cell line producing an unlimited supply of a single, chemically defined antibody — the reagent on which modern diagnostics and a large part of modern therapeutics depend.

HISTORICAL ANCHOR — THE 1984 NOBEL PRIZE

Niels K. Jerne, Georges J. F. Köhler and César Milstein, “for theories concerning the specificity in development and control of the immune system **and the discovery of the principle for production of monoclonal antibodies**”. Köhler and Milstein published the hybridoma method in 1975 and **did not patent it**. Every therapeutic monoclonal on the market descends from that decision.

Engineered antibodies — and how to read their names

NAME DECODER — THERAPEUTIC MONOCLONAL ANTIBODIES

-momab	Fully mouse	<i>Most immunogenic. Largely abandoned</i>
-ximab	Chimeric — mouse V, human C	<i>rituximab, infliximab</i>
-zumab	Humanised — only the CDRs are mouse	<i>trastuzumab, bevacizumab</i>
-umab	Fully human	<i>adalimumab, nivolumab</i>
-cept	Receptor-Fc fusion, not an antibody	<i>etanercept — a TNF receptor stuck onto an IgG Fc</i>

Reading it: the suffix tells you how much of the molecule is mouse, and therefore how likely the patient is to raise **anti-drug antibodies** against it and lose the response. The progression *momab* → *ximab* → *zumab* → *umab* is the history of the field in four syllables: each step replaced more mouse sequence with human, and the whole point was to keep the mouse **CDRs** — the part that does the binding — while discarding the mouse framework and constant region that the immune system reacts to.

4.2 Antibodies in the Clinic and the Lab

The lecture lists four applications: **diagnosis, prophylaxis, immunotherapy and scientific research**. All four exploit the same property — an antibody will find one molecule in a mixture of thousands and stick to it.

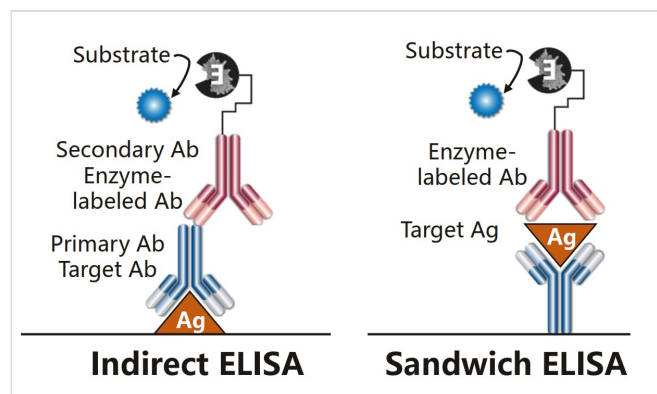


Fig 4.3 ELISA, two formats. **Indirect** — coat the plate with antigen, add patient serum, detect bound antibody with an enzyme-labelled secondary: this asks does the patient have the antibody? **Sandwich** — capture antigen between two antibodies: this asks is the antigen present, and how much?

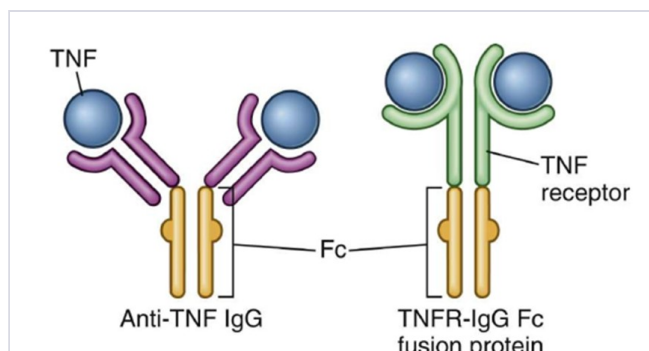


Fig 4.4 Two ways to block one cytokine. **Anti-TNF IgG** binds TNF directly. A **TNFR-IgG Fc fusion protein** instead presents the receptor itself as a decoy, dimerised and given a long half-life by the Fc it is fused to. Both neutralise TNF; only one is an antibody.

Inflammatory (immunological) diseases		
Target	Effect	Diseases
CD20	Depletion of B cells	B cell lymphomas, rheumatoid arthritis, multiple sclerosis, other autoimmune diseases
IgE	Blocking mast cell sensitization	Asthma, peanut allergy
IL-4/IL-13 receptor	Blocking inflammation	Asthma, atopic dermatitis
IL-5	Blocking eosinophil activation	Asthma
IL-6 receptor	Blocking inflammation	Rheumatoid arthritis, cytokine storm
IL-17	Blocking inflammation	Psoriasis
TNF	Blocking inflammation	Rheumatoid arthritis, Crohn's disease, psoriasis
α 4 integrin	Blocking leukocyte migration	Inflammatory bowel disease; multiple sclerosis
C5	Blocking complement-mediated cell lysis	Paroxysmal nocturnal hemoglobinuria, hemolytic uremic syndrome
Cancer		
Target	Effect	Diseases
CD52	Depletion of lymphocytes	Chronic lymphocytic leukemia
CTLA-4	Activation of T cells	Melanoma, lung cancer
EGFR	Growth inhibition of epithelial tumors	Colorectal, lung, and head and neck cancers
HER2/NEU	Inhibition of EGF signaling; depletion of tumor cells	Breast cancer
PD-1, PD-L1	Activation of effector T cells	Many tumors
VEGF	Blocking tumor angiogenesis	Breast cancer, colon cancer, age-related macular degeneration
Other diseases		
Target	Effect	Diseases
Glycoprotein IIb/IIIa	Inhibition of platelet aggregation	Cardiovascular disease
SARS-CoV2 spike protein	Blocking SARS-CoV2 infection of cells	COVID-19

Fig 4.5 Therapeutic monoclonals in use, by target and disease. Read the first column as a list of the molecules immunology has identified as worth blocking — the table is, in effect, immunology's results made into drugs.

CLINICAL CORRELATION — THE 2018 NOBEL PRIZE AND CHECKPOINT BLOCKADE

James P. Allison and Tasuku Honjo won the 2018 Nobel Prize “for their discovery of cancer therapy by inhibition of negative immune regulation.” Read that phrase carefully, because it describes an inversion. Earlier cancer immunotherapy tried to **push the accelerator** — give cytokines, give vaccines, stimulate T cells. Allison and Honjo showed that tumours survive by **engaging the brakes** that normally stop T cells attacking self: CTLA-4 and PD-1. The therapy is therefore a **monoclonal antibody that blocks a brake**, not one that stimulates anything — ipilimumab against CTLA-4, nivolumab and pembrolizumab against PD-1. The side effects follow directly and predictably: release the brakes on self-tolerance and you get **autoimmune** colitis, thyroiditis, hepatitis and dermatitis. The mechanism predicts the toxicity, which is the sign that the mechanism is right.

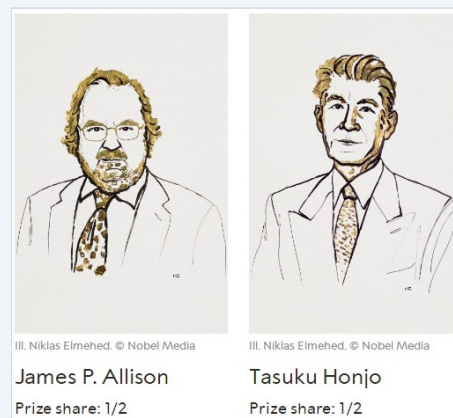


Fig 4.6 James P. Allison and Tasuku Honjo, Nobel Prize **2018** — the reason a chapter on antibody engineering now ends in oncology.

05

Complement — The Cast and the Three Pathways

Thirty-odd proteins, but only one event that matters: **the cleavage of C3**. Three pathways are three different ways of building the enzyme that does it, and after that point all three are the same. Learn it in that shape and the cast list stops being a list.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Read any complement symbol you meet, including ones not taught here.
- Name the initiator, the C3 convertase and the C5 convertase of each pathway.
- Explain why free antibody cannot activate complement but bound antibody can.
- Describe how the MAC kills a cell.

5.1 What Complement Is, and How to Read Its Names

DEFINITION

The **complement system** consists of a **large number of proteins that react with one another to opsonise pathogens** and to **induce a series of inflammatory responses** that help to fight infection.

HISTORICAL ANCHOR — WHERE THE NAME COMES FROM

Jules Bordet (1870–1961, Nobel Prize 1919) discovered complement as a **heat-labile component of normal plasma that augments the opsonisation and killing of bacteria by antibodies**. This activity was said to *complement* the antibacterial activity of antibody — hence the name. Note what that history tells you: complement was found as **antibody's helper**, which is why the antibody-dependent route is called the *classical* pathway even though it was, in evolutionary terms, the last to appear.

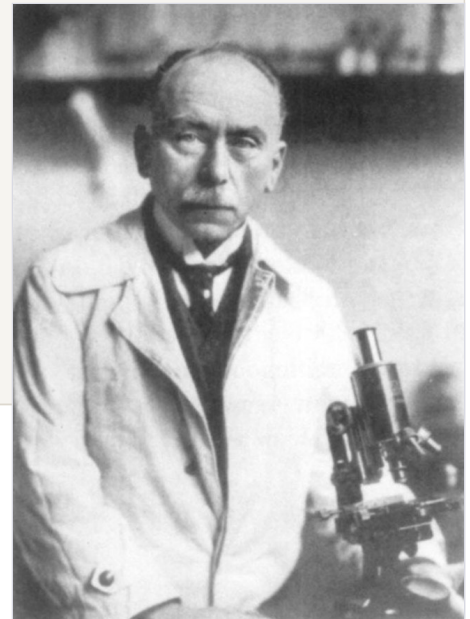


Fig 5.1 Jules Bordet (1870–1961), Nobel Prize 1919. He found complement by heating serum: the antibacterial power of fresh serum disappeared at 56 °C, and came back when fresh serum was added. Something heat-labile was helping the antibody — and that something is this whole Part.

NAME DECODER — COMPLEMENT NOTATION

C	Complement component	<i>C1, C2, C3, C5</i>
a	The small fragment after cleavage	<i>C3a, C5a — these diffuse away and act as inflammatory mediators</i>
b	The large fragment after cleavage	<i>C3b, C4b, C5b — these stay attached to the surface and build the next enzyme</i>
$\bar{C}$ (overbar)	Activated — it is now an enzyme	<i>$\bar{C}1$ means C1 that has become proteolytically active</i>
i	Inactivated	<i>iC3b — C3b that factor I has cut up; still an opsonin, no longer an enzyme</i>
Letters	Alternative-pathway proteins have letters, not numbers	<i>Factor B, Factor D, Factor H, Factor I, properdin</i>

Worked example — C4b2a. Read it left to right as a list of assembled parts: **C4b** (the large fragment of C4, stuck to the surface) plus **2a** (a fragment of C2) — two pieces joined into one complex. Because it is a complex of activated fragments sitting on a membrane, it is an **enzyme**, and the fact that it appears in the classical pathway immediately after C4 and C2 tells you what it must cleave next: **C3**. So C4b2a is the **classical-pathway C3 convertase**, and you have derived that rather than memorised it. Add another C3b — **C4b2a3b** — and the specificity shifts to C5.

EXAM TRAP — C2A OR C2B?

C2 is **the one exception** to the a-is-small rule, and textbooks disagree. In the older convention used by this lecture, the **large** fragment of C2 is called **C2a** and the convertase is written **C4b2a**; some modern texts reverse it and write C4b2b. **Follow your lecturer's convention** — here, C4b2a. If a question gives you the alternative, recognise that it means the same complex and do not treat it as a different molecule. The underlying biology never changes: the large fragment of C2 joins C4b to make the classical C3 convertase.

Properties of complement

TABLE 5.1 WHAT KIND OF SUBSTANCE COMPLEMENT IS

Property	And why it matters
Glycoprotein — globulins	Present in normal serum at all times , before any infection. Complement is not induced by the pathogen; it is already there, waiting
Heat-labile	Activity is destroyed by heating serum to 56 °C for 30 minutes . This is a real laboratory procedure — “heat-inactivated serum” means exactly this, and it is done routinely so that complement in the serum does not interfere with an assay
Produced by hepatocytes and macrophages	Mostly the liver. Severe liver disease therefore lowers complement — one more reason cirrhotic patients get infections
Acute-phase protein	Its concentration rises during inflammation , driven by IL-6 acting on the liver. So complement levels are <i>higher</i> in acute inflammation and <i>lower</i> when it is being consumed by immune complexes — a distinction that matters clinically (§6.4)

5.2 The Cast List

Functional protein classes in the complement system		Functional protein classes in the complement system	
Binding to antigen:antibody complexes and pathogen surfaces	C1q	Membrane-attack proteins	C5b C6 C7 C8 C9
Binding to mannose on bacteria	MBL		
Activating enzymes	C1r C1s C2b Bb D MASP-1 MASP-2	Complement receptors	CR1 CR2 CR3 CR4 C1qR
Membrane-binding proteins and opsonins	C4b C3b	Complement-regulatory proteins	C1INH C4bp CR1 MCP DAF H I P CD59
Peptide mediators of inflammation	C5a C3a C4a		

Figure 2-20 Immunobiology, 6/e. (© Garland Science 2005)

Fig 5.2 The whole system sorted by **job** rather than by pathway: proteins that bind antigen–antibody complexes or microbes, activating enzymes, membrane-binding and opsonic proteins, peptide mediators of inflammation, complement receptors, and complement-regulatory proteins. Six jobs — every protein in the system does one of them.

TABLE 5.2 THE THREE GROUPS OF COMPLEMENT PROTEIN

Group	Members
1. Innate components — the cascade itself	Classical CP : C1q, C1r, C1s, C4, C2, C3 · MBL / lectin LP : MBL, MASP, and then C4, C2, C3 · Alternative AP : factor B, factor D, C3 · Terminal TP : C5, C6, C7, C8, C9
2. Control proteins — the brakes	C1-INH · C4-binding protein (C4BP) · factor I · factor H · S protein (vitronectin) · Sp40/40 (clusterin) · decay-accelerating factor (DAF, CD55) · membrane cofactor protein (MCP, CD46) · MIRL (CD59)
3. Receptors — how cells read the signal	CR1 to CR5 · C3aR · C4aR · C5aR

HIGH-YIELD — THE SHAPE OF THE WHOLE SYSTEM IN ONE SENTENCE

Three pathways converge on one enzyme (a C3 convertase), which cleaves C3 into C3a and C3b; C3b then does the opsonising and builds the C5 convertase, which starts the terminal pathway that builds the MAC. Everything else in the cast list is either a component of one of those steps, a receptor that reads its output, or a brake that stops it going too far. If you can say that sentence you have the architecture, and the details have somewhere to attach.

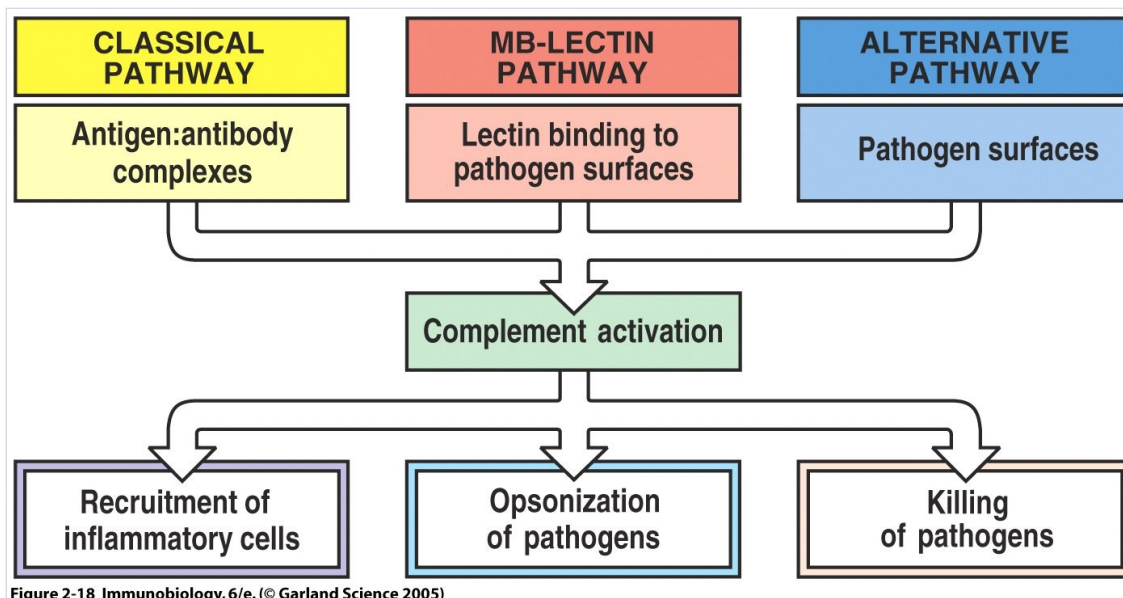
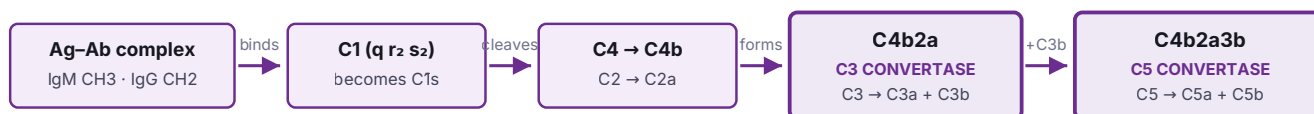


Fig 5.3 The same architecture as a diagram. Three different recognition events at the top — **antigen–antibody complexes**, **lectin binding to pathogen surfaces**, **pathogen surfaces directly** — funnel into **complement activation**, and out of the bottom come the same three outputs every time: **recruitment of inflammatory cells**, **opsonisation**, **killing**.

5.3 The Classical Pathway



RECOGNITION → ACTIVATION → TERMINAL

The lecture divides it in exactly these three stages. C1 activation is recognition; C4, C2 and C3 are activation; C5 onwards is the terminal effect.

Fig 5.4 The classical pathway as a strip. Read each arrow as “acts on”. Note that **C4 comes before C2** despite the numbering — the components were numbered in order of discovery, not order of action.

PATHWAY 1 OF 3 · CP

The classical pathway

WHAT STARTS IT

The antigen–antibody complex (immune complex) — not antibody alone.

C1q binds **CH2 of IgG** (IgG1, IgG2, IgG3) or **CH3 of IgM**.

One IgM molecule is enough; **two IgG molecules** are needed.

Soluble antibody cannot activate complement.

WHY FREE ANTIBODY CANNOT DO IT

C1q has **six globular heads** on six collagen-like stalks, and it must engage **at least two heads simultaneously** to change shape and activate C1r and C1s.

Because: In free solution, IgG molecules are far apart and their Fc regions point in random directions, so two heads can never be engaged at once. The moment the antibodies bind a surface, the geometry changes: they are **fixed, close together and all pointing the same way**, so C1q can straddle two adjacent Fc regions. This is why complement activation is **automatically restricted to real targets** — the system needs no separate mechanism to avoid attacking free antibody, because free antibody is physically incapable of triggering it. For IgM the logic is the same but easier: pentameric IgM already carries five clustered Fc regions, and on binding antigen it flips from a flat “star” to a “staple” conformation that exposes the C1q sites which were hidden before.

KEY COMPLEXES

C3 convertase = C4b2a · C5 convertase = C4b2a3b

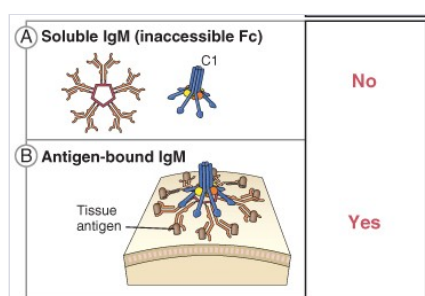


Fig 5.5 IgM. Soluble pentamer — the Fc portions are **inaccessible**, answer **No**. Antigen-bound — the molecule bends into the staple form and the C1q sites are exposed, answer **Yes**.

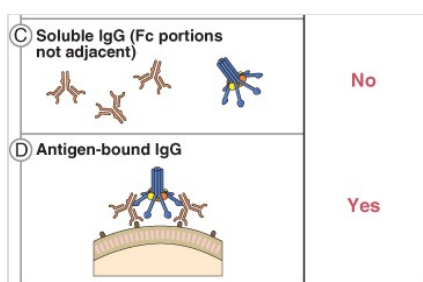


Fig 5.6 IgG. Soluble — the Fc portions are **not adjacent**, answer **No**. Bound to a tissue antigen — two IgG land close enough together for C1q to bridge them, answer **Yes**.

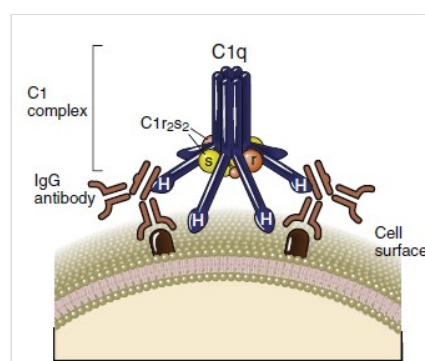
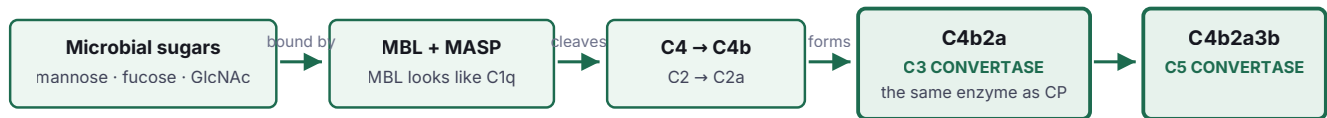


Fig 5.7 The C1 complex: one **C1q** — the six-headed bunch of tulips — plus **C1r2s2** held in its arms. The heads reach down to the Fc regions of antibody already bound to the cell surface.

5.4 The Lectin (MBL) Pathway



Identical to the classical pathway from C4 onwards. **The only difference is what does the recognising: a lectin instead of an antibody.**

Fig 5.8 The lectin pathway. Compare with Fig 5.4: from the third box onwards the two strips are the same.

PATHWAY 2 OF 3 · LP

The mannan-binding lectin pathway

WHAT STARTS IT

MBL recognises certain carbohydrates — mannose, fucose and GlcNAc — expressed on the surface of microorganisms.

MBL is like C1q in structure — the same bunch-of-tulips architecture, which is why it can do the same job.

MBL is produced by hepatocytes in the acute-phase reaction.

MASP (MBL-associated serine protease) takes the place of C1r and C1s.

WHY IT EXISTS AT ALL

It is the **classical pathway without the antibody** — and therefore without the wait.

Because: *The classical pathway is fast once you have antibody, but making antibody against a new pathogen takes **days**. The lectin pathway gets the same machinery started **immediately**, because MBL recognises a pattern rather than a specific epitope. The pattern it recognises is the key point: **terminal mannose and fucose in the arrangement found on microbes, not on human cells**. Human glycoproteins terminate in sialic acid and galactose instead, so MBL ignores them. The system distinguishes self from non-self by sugar chemistry — no learning, no memory, no antibody required. That is innate immunity working exactly as advertised.*

KEY COMPLEXES

C3 convertase = C4b2a · C5 convertase = C4b2a3b — identical to the classical pathway

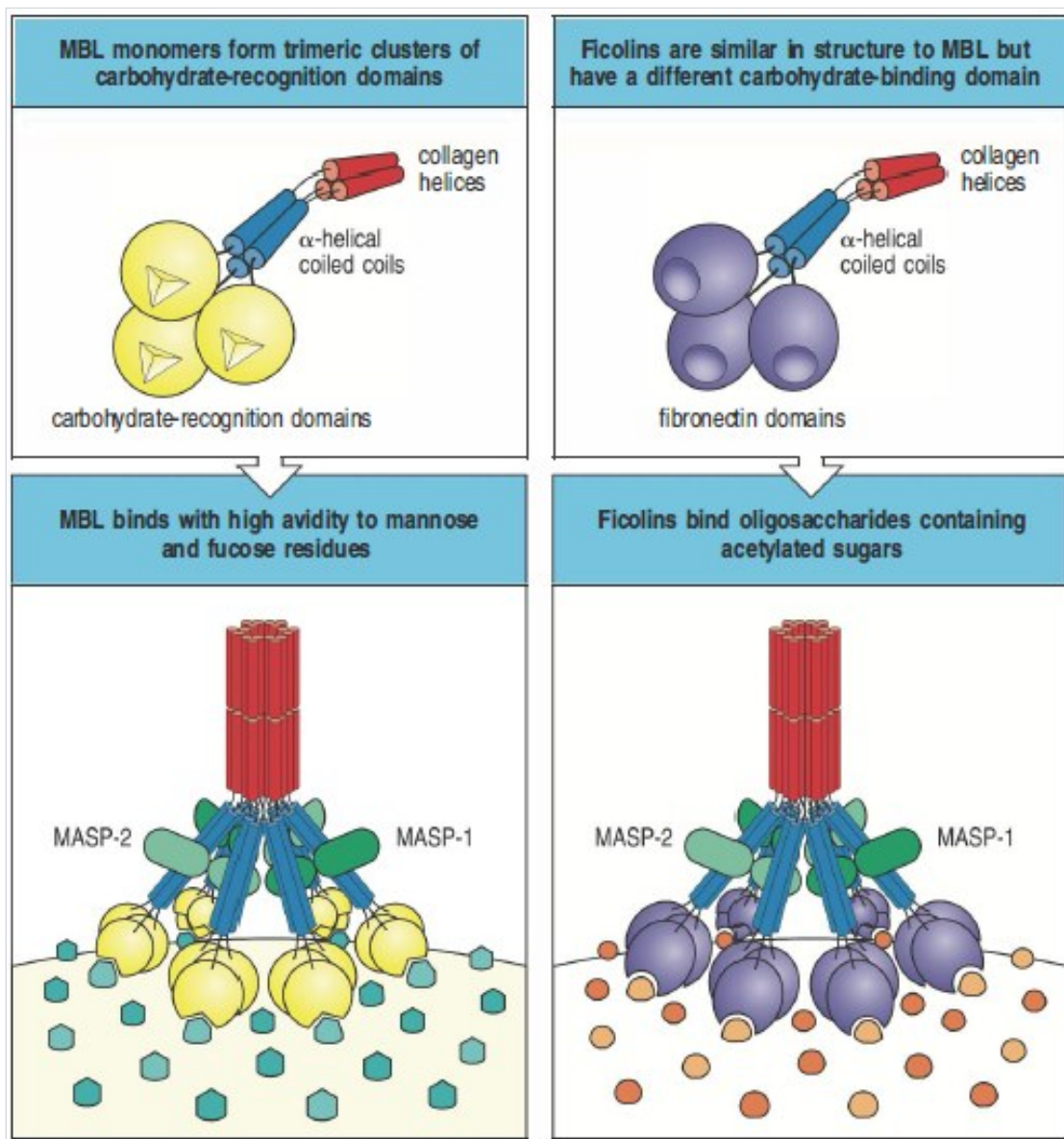


Fig 5.9 MBL (left) and ficolins (right), both with MASPs attached, binding the repeating sugar arrays on a bacterial surface. Note how similar the architecture is to C1q in Fig 5.7 — a collagenous stalk carrying multiple recognition heads. Evolution built the classical pathway by borrowing this design and swapping the heads for antibody-binding ones.

5.5 The Alternative Pathway

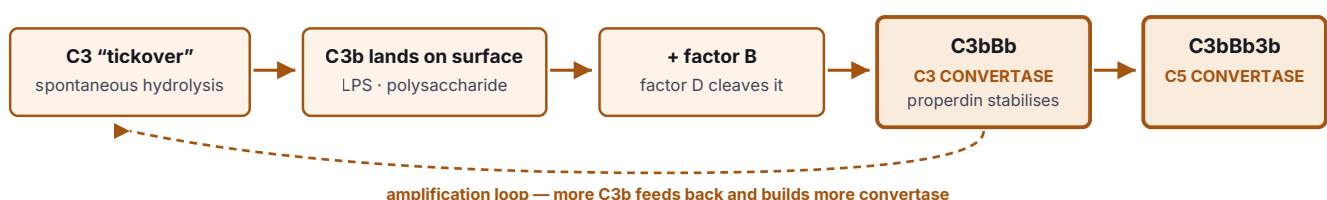


Fig 5.10 The alternative pathway, with its defining feature: the product of the enzyme is also the **substrate for building more of the enzyme**. Note it uses **letters**, not C4 and C2.

PATHWAY 3 OF 3 · AP

The alternative pathway**WHAT STARTS IT**

LPS (endotoxin) and **polysaccharides** — microbial surfaces, directly.

No antibody and no lectin required. Also activated by aggregated IgA.

Components have **letters**: factor B, factor D, properdin — plus C3.

HOW IT CAN START WITHOUT ANY TRIGGER AT ALL

It is **always running at a low level**, and the “decision” is made afterwards.

*Because: C3 in plasma **hydrolyses spontaneously** at a slow rate — the **tickover** — generating a trickle of C3b continuously. That C3b attaches indiscriminately to whatever surface is nearest, including your own cells. What distinguishes friend from foe is not the landing but **what happens next**: your own cells carry regulators — DAF, MCP, factor H — that immediately dismantle any convertase that starts to form (§6.2). A bacterial surface carries none of them, so there the convertase survives, and the **amplification loop** takes over: each C3b made builds another convertase, which makes more C3b. Within minutes a bacterium is coated in thousands of C3b molecules while your erythrocyte beside it is untouched. **The alternative pathway does not recognise the pathogen — it fails to be switched off on it.***

KEY COMPLEXES

C3 convertase = C3bBb · C5 convertase = C3bBb3b — note there is **no C4** and **no C2** anywhere in this pathway

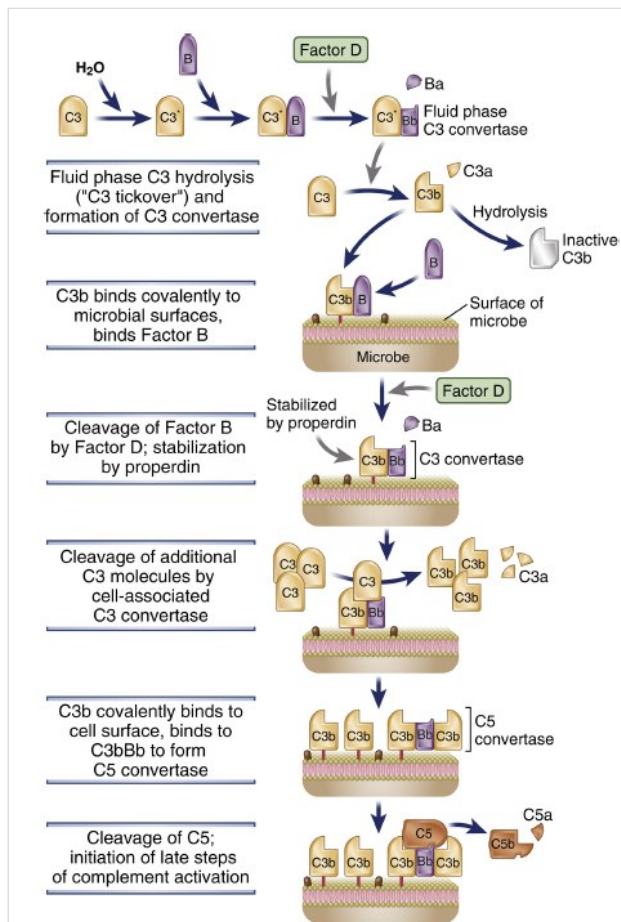


Fig 5.11 The alternative pathway drawn step by step. Read it downwards: spontaneous hydrolysis of C3 in the fluid phase (tickover) → factor B binds → factor D cleaves it → properdin stabilises the resulting C3bBb on the microbial surface → more C3 is cleaved → a second C3b joins to give C3bBbC3b, the C5 convertase. On a host cell the same tickover happens, but regulators dismantle it before step three.

5.6 The Terminal Pathway and the MAC

From C5 onwards there is only one pathway. Whichever convertase cleaved C5, what follows is identical — and unlike everything before it, this stage involves **no enzymes at all**. It is pure self-assembly.

MECHANISM — HOW FIVE PROTEINS PUNCH A HOLE IN A MEMBRANE

- 1 A C5 convertase cleaves C5 → C5a + C5b.**
WHY: C5a floats away and becomes the most potent anaphylatoxin and chemotaxin in the system (§6.3). C5b stays on the surface and is the seed for everything that follows.
- 2 C6 and C7 bind C5b in turn, and C7 exposes a hydrophobic site.**
WHY: This is the committing step. Until now everything has been water-soluble; C5b67 can insert into the **lipid bilayer**. If it does not find a membrane quickly it is neutralised in the fluid phase — which is what S protein does (§6.2).
- 3 C8 binds and pushes further into the bilayer, making a small unstable pore.**
WHY: C8 alone leaks slightly. The damage so far is survivable.
- 4 C8 recruits C9, and C9 polymerises — 10 to 16 molecules of C9 forming a ring.**
WHY: This is the amplification of the terminal pathway: one C8 nucleates many C9. The ring is the **membrane attack complex (MAC)**, and its channel is about 10 nm across — large enough to let ions and small molecules pass freely.

- 5 The cell can no longer maintain its ionic gradients. **Water enters down its osmotic gradient and the cell lyses.**

WHY: The MAC does not digest the cell or poison it — it simply removes the membrane's ability to hold a gradient, and osmosis does the rest. This is why a bacterium with a thick Gram-positive peptidoglycan wall is **resistant** to MAC: the complex cannot reach the membrane through the wall. It is also why **Neisseria**, which is Gram-negative and thin-walled, is the one organism against which the MAC is indispensable.

NET RESULT A ring-shaped transmembrane channel — and a cell that dies by **osmotic lysis**, not by anything the complement proteins did chemically.

The terminal complement components that form the membrane-attack complex		
Native protein	Active component	Function
C5	C5a	Small peptide mediator of inflammation (high activity)
	C5b	Initiates assembly of the membrane-attack system
C6	C6	Binds C5b; forms acceptor for C7
C7	C7	Binds C5b6; amphiphilic complex inserts in lipid bilayer
C8	C8	Binds C5b67; initiates C9 polymerization
C9	C9 _n	Polymerizes to C5b678 to form a membrane-spanning channel, lysing cell

Figure 2-34 Immunobiology, 6/e. (© Garland Science 2005)

Fig 5.12 The terminal components with their serum concentrations and functions — **C5b** initiates assembly, **C6** binds C5b, **C7** gives lipid affinity, **C8** starts pore formation, and **C9** polymerises to complete the channel.

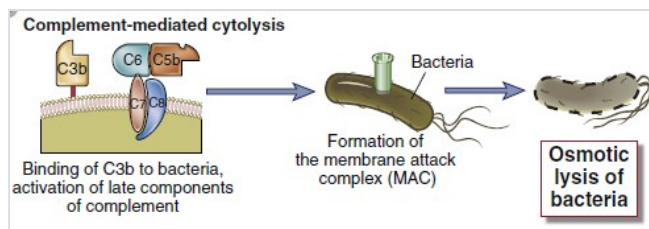


Fig 5.13 The result: C5b–C9 assembles on the bacterial membrane, forms the MAC, and the organism dies by **osmotic lysis**.

06

Complement – Regulation, Function, Disease

Part 5 asked how complement is switched on. This Part asks the harder question – **why is it not on right now, everywhere, in you?** Complement is a self-amplifying proteolytic cascade running continuously in your plasma, and the only reason it is not dissolving your own cells is a second system of proteins whose whole job is to restrain it. Learn the brakes and the functions fall out of them; learn the brakes and every complement disease in the syllabus becomes a brake that failed.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Explain why complement needs regulation at all, and name the three points where regulators act.
- Match every regulator to its target, its mechanism and the disease of its absence.
- List the six biological functions and name the fragment and receptor responsible for each.
- Explain why C3 deficiency is the most severe and why terminal-component deficiency means *Neisseria*.
- Distinguish hereditary angioedema from allergic angioedema on mechanism alone.

6.1 Why Complement Must Be Restrained

Everything in Part 5 was about getting the cascade started. Start it and you have a problem, because of two properties that Part 5 deliberately did not dwell on.

THE TWO PROPERTIES THAT MAKE REGULATION COMPULSORY

1. Amplification. A single C3 convertase sitting on a surface cleaves **thousands** of C3 molecules. In the alternative pathway the product of the enzyme (C3b) is also a building block of the enzyme, so the output of one round is the input of the next. Nothing in the cascade itself limits the total.

2. C3b is chemically, not biologically, targeted. When C3 is cleaved, a buried **thioester bond** is exposed for a few microseconds and reacts with any hydroxyl or amino group within reach. That is a *chemical* reaction. It does not ask whose surface it is. C3b will attach to a bacterium, to a red cell, to your glomerular basement membrane — whatever is nearest.

Add to that the **tickover** of \$5.5 — C3 hydrolysing spontaneously in plasma all day long — and you arrive at the central insight of this Part.

THE CENTRAL INSIGHT

Complement does **not** discriminate self from non-self by recognising the enemy. It attacks everything, and **self is protected**. A host cell is spared not because complement fails to land on it, but because the host cell carries **regulatory proteins on its membrane** that dismantle the cascade within seconds. A bacterium has no such proteins — and that, and only that, is the difference.

REMEMBER IT AS

Complement is a fire, not a sniper. It burns whatever it touches. Your cells are fireproofed; microbes are not.

Three places to put a brake

Look back at the cascade and there are only three moments where intervention is possible: before the convertase is built, while the convertase exists, and after C5b has been released. Every regulator in the syllabus sits at one of these three, and knowing **which one** is usually the whole exam question.

TABLE 6.1 THE THREE CONTROL POINTS

Control point	What is stopped	Who does it
1. Recognition — before anything is cleaved	The initiating enzyme is neutralised while it sits on the antibody or the lectin	C1-INH (inhibits C1r ₂ S ₂ and the MASPs)
2. The C3/C5 convertase — the amplification step	Either the convertase is pulled apart (decay), or its C3b/C4b core is cut up by factor I so it can never be rebuilt	DAF/CD55, CR1/CD35, C4bp, factor H (decay) · MCP/CD46, CR1, C4bp, factor H as cofactors for factor I (cleavage)

Control point	What is stopped	Who does it
3. The MAC — after C5b is free	C5b67 is either trapped in the fluid phase before it can insert, or C9 is prevented from polymerising once it has	S protein (vitronectin), clusterin/Sp40,40 (fluid phase) · CD59/MIRL (on the membrane)

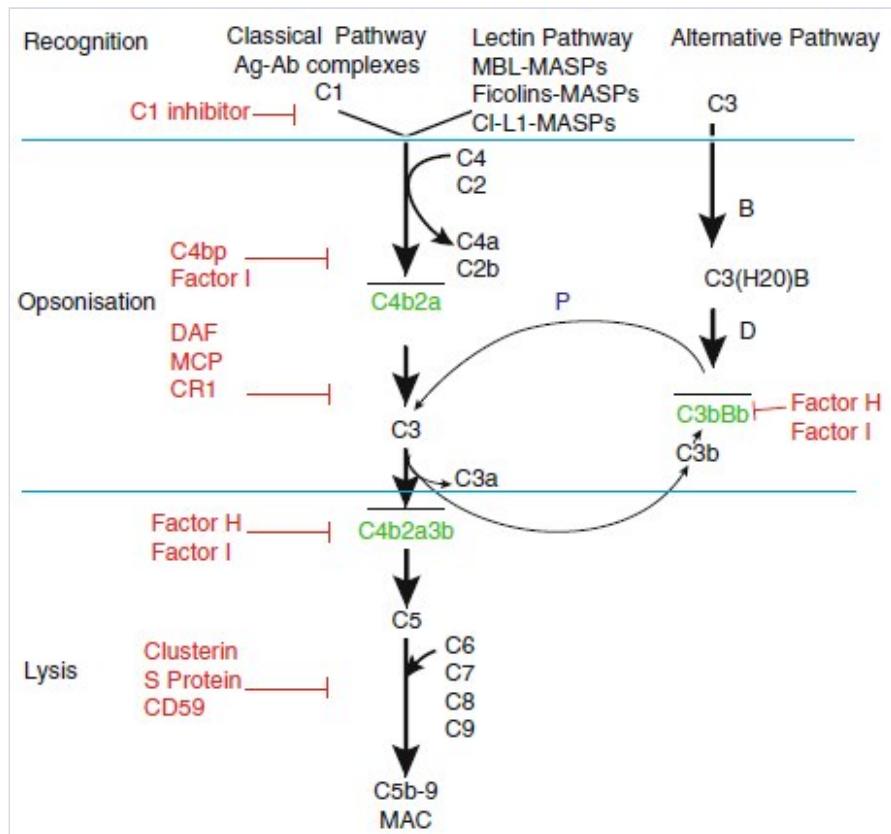


Fig 6.1 The single most useful diagram in this Part. The three pathways run down the page in black; every regulator is in red, drawn as a **bar across the arrow it blocks**. Read it by the three horizontal bands on the left — **recognition** (C1 inhibitor), **opsonisation** (C4bp, factor I, DAF, MCP, CR1, factor H) and **lysis** (clusterin, S protein, CD59). Notice that the densest cluster of brakes is exactly at the C3 convertase, because that is where the amplification is.

6.2 The Regulators, One by One

There are three mechanisms and three mechanisms only. Once you can say which of the three a protein uses, you have understood it.

NAME DECODER — THE THREE VERBS OF COMPLEMENT REGULATION

Decay	Pull the convertase apart	<i>Bind the C4b or C3b and physically displace C2a or Bb. The convertase falls off; the C4b/C3b is still there but is now inert</i>
Cofactor	Hold it still for the knife	<i>Factor I is the only protease that destroys C3b and C4b, but it cannot work alone. A cofactor binds C3b first and presents it. Result: C3b → iC3b — permanent, irreversible</i>
Block	Stop the pore	<i>Either grab C5b67 while it is still soluble (S protein, clusterin) or sit in the membrane and refuse to let C9 polymerise (CD59)</i>
Serpin	Trap the protease	<i>A suicide-substrate inhibitor. C1-INH is the only one here, and it acts upstream of everything else</i>

Worked example — why the same job needs both DAF and MCP. DAF *decays*: it knocks C2a off C4b, which stops this convertase immediately but leaves C4b on the membrane, free to recruit a new C2 and start again. MCP is the *cofactor*: it hands that C4b to factor I, which cuts it into fragments that can never bind C2 again. **DAF buys time; MCP makes it permanent.** That is why a cell carries both, and why losing either one is not the same disease.

TABLE 6.2 THE COMPLEMENT REGULATORY PROTEINS — THE MASTER TABLE

Regulator	Soluble or membrane	Target	Mechanism	When it fails
C1-INH C1 esterase inhibitor	Soluble	C1r ₂ S ₂ and MASP-1/2 CP LP	Serpin. Binds the active site of C1r and C1s and drags them off C1q, so the classical pathway never starts. Does the same to the MASPs	Hereditary angioedema — and the mechanism is <i>not</i> complement (\$6.4)
C4bp C4-binding protein	Soluble	C4b CP LP	Both. Decays C4b2a and acts as cofactor for factor I on C4b	Uncontrolled classical/lectin amplification
Factor H	Soluble	C3b AP	Both — and it is the protein that makes the alternative pathway self-tolerant. It binds sialic acid and heparan sulphate , which host cells carry and most bacteria do not. So factor H docks on <i>your</i> membranes, decays C3bBb there and delivers C3b to factor I; on a bacterium it cannot dock, and the loop runs free	Atypical HUS, C3 glomerulopathy, age-related macular degeneration — all diseases of an alternative pathway that will not switch off
Factor I	Soluble	C3b and C4b — all pathways	The knife. A serine protease that cleaves C3b to iC3b (and on to C3d) and C4b to C4c + C4d. Absolutely requires a cofactor. This is the only irreversible step in the whole regulatory system	C3 is consumed continuously → a secondary C3 deficiency with severe pyogenic infection

Regulator	Soluble or membrane	Target	Mechanism	When it fails
DAF / CD55 decay-accelerating factor	Membrane <i>GPI-anchored</i>	C4b2a and C3bBb	Decay only. Displaces C2a and Bb from convertases forming on a host membrane. Works on all three pathways because all three converge on a C3 convertase	With CD59, lost in PNH — see below
MCP / CD46 membrane cofactor protein	Membrane	C3b, C4b	Cofactor only. Presents C3b/C4b on the host membrane to factor I, converting it to iC3b	Atypical HUS
CR1 / CD35	Membrane	C3b, C4b	Both — decay <i>and</i> cofactor. It is also the receptor that erythrocytes use to carry immune complexes to the liver (§6.3), so the same molecule does regulation and clearance	Low CR1 on red cells is found in SLE — immune complexes are not cleared
S protein (vitronectin) Clusterin (Sp40,40)	Soluble	C5b67 TP	Block, fluid phase. Binds the nascent C5b67 complex while it is still in solution and masks its hydrophobic site, so it can never insert into a membrane	Bystander lysis of innocent cells next to the target
CD59 / MIRL membrane inhibitor of re-active lysis	Membrane <i>GPI-anchored</i>	C5b678 TP	Block, on the membrane. Binds C8 and C9 in the assembling complex and prevents C9 polymerisation . The MAC stalls before it becomes a channel	With DAF, lost in PNH → intravascular haemolysis

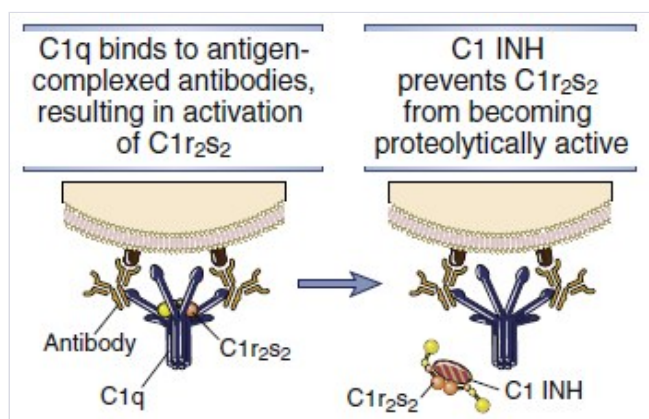


Fig 6.2 C1-INH. Left: C1q binds antigen-complexed antibody and C1r₂s₂ becomes active. Right: C1-INH binds C1r₂s₂ and **drags it off the C1q stalks** before it can cleave anything.

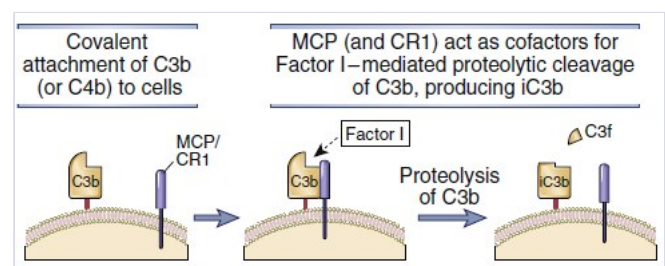


Fig 6.3 MCP/CR1 + factor I. C3b attaches covalently to a host cell (left). MCP or CR1 presents it to **factor I**, which cuts it, releasing the small fragment **C3f** and leaving **iC3b** — still an opsonin, no longer able to build a convertase.

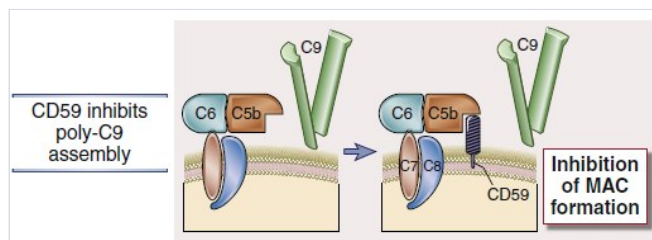


Fig 6.4 CD59 (MIRL). C9 is about to polymerise onto C5b678; CD59 in the host membrane inserts itself between them, and MAC formation is **inhibited**.

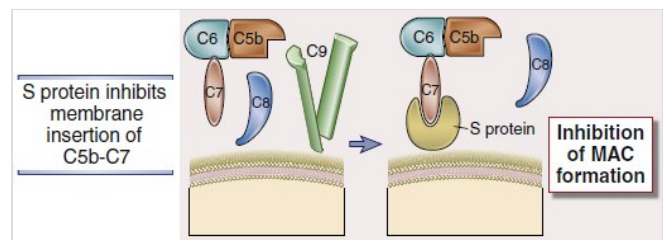


Fig 6.5 S protein (vitronectin). The same endpoint by a different route — C5b67 is capped **in the fluid phase**, so it never reaches the bilayer at all. Compare the two figures: CD59 defends one particular cell; S protein protects every cell in the neighbourhood.

— ABSENCE PROVES FUNCTION — PAROXYSMAL NOCTURNAL HAEMOGLOBINURIA (PNH)

An acquired mutation in the **PIGA** gene of a haematopoietic stem cell destroys the ability to make the **GPI anchor**. Every protein that reaches the membrane by that anchor is therefore missing from the entire clone — and two of them are **DAF (CD55)** and **CD59**. Nothing else about complement is abnormal: C3 is normal, all three pathways are normal, tickover is normal. The red cells are simply no longer fire-proofed, so the everyday, physiological tickover of the alternative pathway lyses them. The result is **chronic intravascular haemolysis** with haemoglobinuria, and thrombosis.

Read what this proves. It proves that complement is attacking your red cells every single day and that CD55 and CD59 are the only reason you do not notice. It also predicts the treatment, and the prediction is correct: **eculizumab**, a monoclonal antibody against **C5** (§4.2, Fig 4.5), blocks the terminal pathway and stops the haemolysis.

EXAM TRAP — DAF, MCP, CD59: WHICH ONE, WHICH STEP?

All three are **membrane** proteins on host cells, all three protect self, and questions routinely swap them. Fix them by the step, not the name:

- **DAF/CD55** → the **convertase**, decay. Acts *before* C5 is ever cleaved.
- **MCP/CD46** → the **C3b itself**, cofactor for factor I. Same stage as DAF, different verb.
- **CD59/MIRL** → the **MAC**, blocks C9. Acts *after* everything upstream has already succeeded.

And one more that is often missed: **C4bp** and **factor H** are the **soluble versions of the same two jobs** — C4bp does for C4b what DAF+MCP do, and factor H does it for C3b. Membrane regulators protect the cell they sit on; soluble regulators patrol the plasma.

6.3 The Six Functions of Complement

The lecture lists six. Five of them are performed by fragments you have already met, and the sixth explains why complement, coagulation and the kinin system keep turning up in the same diseases.

TABLE 6.3 THE SIX FUNCTIONS AT A GLANCE

#	Function	The molecule that does it	Receptor / target	Result
1	Cytolysis	C5b–C9 (MAC)	The lipid bilayer itself — no receptor	Osmotic lysis. Mainly Gram-negative bacteria and enveloped viruses
2	Opsonisation	C3b, C4b, iC3b	CR1, CR3, CR4 on phagocytes	Enhanced phagocytosis. Quantitatively the most important antibacterial function of complement
3	Inflammation	C5a, C3a, C4a — the anaphylatoxins	C5aR, C3aR, C4aR on mast cells, neutrophils, monocytes, smooth muscle	Mast-cell degranulation, vascular permeability, smooth-muscle contraction; C5a is also the chemotaxin
4	Immune-complex clearance	C3b deposited on the complex	CR1 on erythrocytes	Complexes are ferried to liver and spleen and stripped off by macrophages

#	Function	The molecule that does it	Receptor / target	Result
5	Mediating adaptive immunity	C3d (and iC3b)	CR2/CD21 on B cells and follicular dendritic cells	Lowers the B-cell activation threshold; antigen retention for memory
6	Cross-talk with other enzyme systems	Shared proteases and shared inhibitors	Coagulation, fibrinolytic and kinin cascades	One cascade activates the others; this is why a single inhibitor defect causes a non-complement disease

1 & 2 — Cytolysis and opsonisation

Cytolysis is §5.6 and needs nothing added: C5b–C9 makes the MAC and the cell dies osmotically. What is worth stating plainly is the **relative importance**. Lysis only works on cells with an exposed, thin outer membrane — chiefly **Gram-negative bacteria** (especially *Neisseria*) and **enveloped viruses**. A Gram-positive organism has a thick peptidoglycan wall that the MAC cannot reach through, and is essentially resistant to lysis. Against those, complement works by **opsonisation** instead — which is why a C3 deficiency is so much worse than a C9 deficiency.

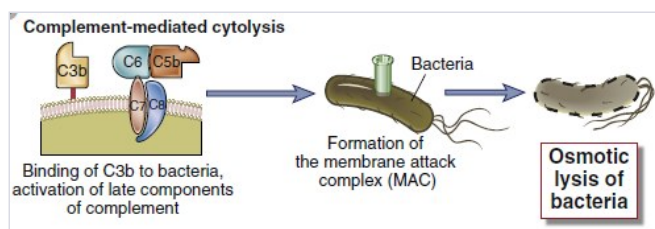


Fig 6.6 Function 1 — cytolysis. C3b binds the bacterium, the late components assemble, the MAC forms, and the organism dies by **osmotic lysis**.

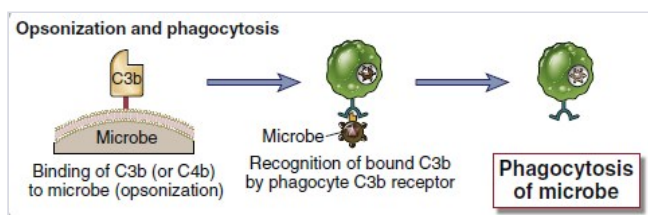


Fig 6.7 Function 2 — opsonisation. C3b (or C4b) coats the microbe; the phagocyte's C3b receptor recognises the coat; the microbe is eaten. Opsonin is Greek for “seasoning” — complement makes the bug taste good.

WHICH OPSONIN, WHICH RECEPTOR

C3b → CR1 (CD35). iC3b → CR3 (CD11b/CD18, Mac-1) and CR4 (CD11c/CD18). This is the reason factor I's product is called *inactivated* C3b and not *destroyed* C3b: cutting C3b to iC3b abolishes its ability to build a convertase but **leaves it a perfectly good opsonin**, now read by a different pair of receptors. Regulation does not switch opsonisation off; it switches amplification off and leaves the flag flying.

3 — Inflammation: the anaphylatoxins

The small fragments — the ones with the letter **a** — do not stay on the surface. They diffuse away into the tissue, and there they are **hormones of inflammation**. Three of them are anaphylatoxins, but only one is also a chemotaxin.

THE POTENCY ORDER, AND THE ONE THAT IS DIFFERENT**C5a > C3a > C4a**

All three are **anaphylatoxins**: they bind their receptors on mast cells and basophils and trigger degranulation, giving histamine release, increased vascular permeability and smooth-muscle contraction — the same clinical picture as anaphylaxis, without any IgE. **C5a alone is also a chemotaxin**, and a powerful one: it is the signal that brings neutrophils and monocytes out of the vessel and along the gradient to the site. If a question asks for *the* complement chemotactic factor, the answer is always **C5a**.

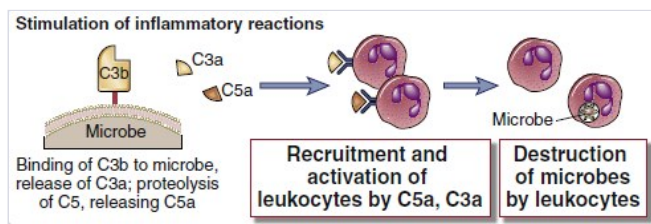


Fig 6.8 Where the anaphylatoxins come from: C3b binds the microbe, **C3a is released**, C5 is then cleaved and **C5a is released**. The two fragments recruit and activate leukocytes, which destroy the microbe.

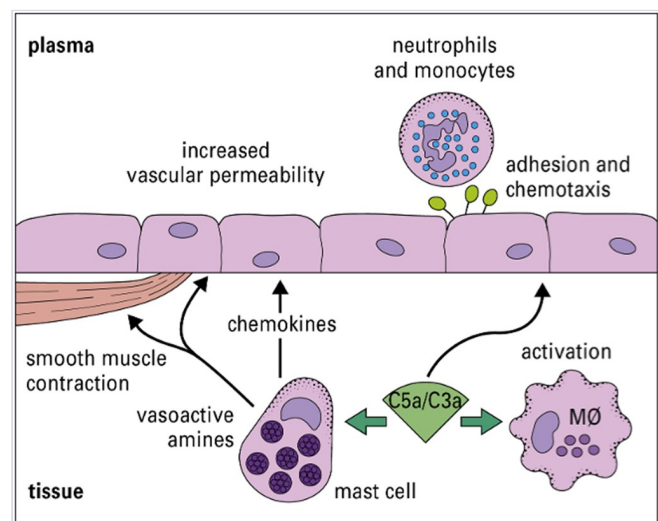


Fig 6.9 The same two fragments seen from the tissue side. **C5a/C3a** act on the **mast cell** (vasoactive amines → increased permeability, smooth-muscle contraction), on the **endothelium** (adhesion molecules and chemokines → neutrophil and monocyte emigration) and on the **macrophage** (activation). Every cardinal sign of acute inflammation is on this one diagram.

4 — Clearing immune complexes**○ MECHANISM — HOW A RED CELL BECOMES A RUBBISH LORRY**

- 1 Antigen and antibody form a **soluble immune complex** in the circulation.

WHY: Soluble is the dangerous state. A complex that stays dissolved travels until it lodges somewhere with high filtration pressure — glomerulus, joint synovium, small vessel wall — and causes disease there.

- 2 The complex activates the **classical pathway**, and **C3b** is deposited all over it.
WHY: *Two things happen at once. The C3b is a handle, and the insertion of C3b between the antibody molecules physically **breaks up the lattice**, keeping the complex small and soluble instead of letting it precipitate.*
- 3 **CR1 on erythrocytes** grips the C3b.
WHY: *Red cells carry far more CR1 in total than all the leukocytes combined, simply because there are so many of them. They are not phagocytes — they are transport.*
- 4 The red cell carries the complex through the circulation to the **liver and spleen**.
- 5 Fixed macrophages there **strip the complex off** and digest it; the erythrocyte is released undamaged and returns to circulation.
WHY: *This is the elegant part — the carrier is not consumed. The same red cell does the job again and again.*

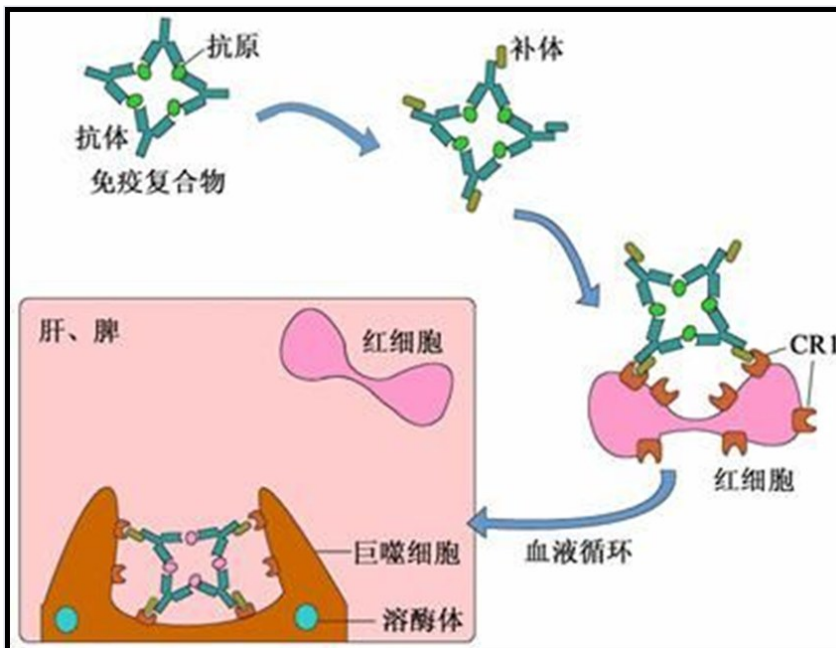


Fig 6.10 Immune-complex clearance. Top left, the complex of antigen (抗原) and antibody (抗体); top right, complement (补体) is deposited on it; right, the complex is gripped by **CR1** on an erythrocyte (红细胞) and carried in the blood (血液循环) to liver and spleen (肝、脾), where a macrophage (巨噬细胞) strips it off into its lysosomes (溶酶体).

CLINICAL CORRELATION — THE COMPLEMENT PARADOX OF SLE

Complement **causes** tissue damage in immune-complex disease — and complement **deficiency causes** immune-complex disease. Both statements are true, and this mechanism is why. If the classical pathway cannot deposit C3b on complexes, they are never handed to erythrocyte CR1, never delivered to the liver, and instead circulate until they precipitate in vessel walls, glomeruli and skin. That is why **deficiency of C1q, C1r, C1s, C4 or C2 produces a lupus-like illness**, and why C1q deficiency has the highest penetrance for SLE of any known genetic defect.

The laboratory consequence. In active SLE, complement is being *consumed* by complexes, so **C3 and C4 fall** — low complement is a marker of disease activity. In ordinary acute inflammation complement is an acute-phase protein and **rises** (§5.1). The same measurement means opposite things, and the difference is whether complement is being made or being eaten.

5 — Mediating adaptive immunity

Complement is usually filed under innate immunity, and the lecture's fifth function is the correction to that. Complement is the bridge: it is how an innate recognition event is converted into a stronger, longer antibody response. The lecture lists five sub-functions.

TABLE 6.4 THE FIVE WAYS COMPLEMENT SERVES ADAPTIVE IMMUNITY

Sub-function	How
1. Induction of the immune response	C3d left on the antigen after factor I has finished with C3b binds CR2 (CD21) on the B cell. CR2 is part of the CD21–CD19–CD81 co-receptor complex, which is physically coupled to the BCR. Engaging both at once lowers the activation threshold by up to 1000-fold — the same antigen now needs a thousand times less of itself to trigger the B cell
2. Proliferation and differentiation of immune cells	The same co-receptor signal drives the B cell into division and towards plasma-cell differentiation
3. The effector stage of the response	Antibody made at the end of the response works largely <i>through</i> complement — opsonisation and lysis are complement functions that antibody merely targets
4. Immune memory	Follicular dendritic cells in the germinal centre hold antigen on their surface <i>via</i> CR2 for months to years. That retained antigen is what selects high-affinity B cells and maintains memory
5. Clearance of antigen	Function 4 above — the erythrocyte–CR1 transport system

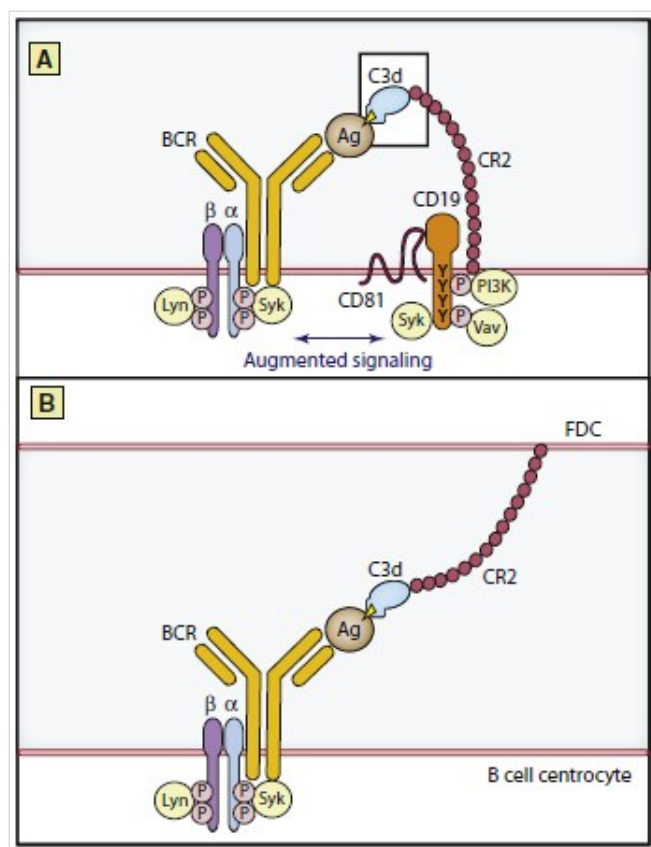


Fig 6.11 A: antigen carrying **C3d** engages the BCR and **CR2** simultaneously; CR2 recruits **CD19**, which is phosphorylated and brings in PI3K and Vav — **augmented signalling**. **B:** the same C3d–CR2 link on a follicular dendritic cell presenting antigen to a centrocyte.

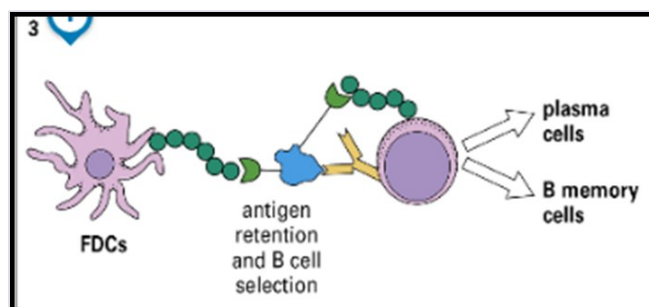


Fig 6.12 Antigen retention and B-cell selection. Follicular dendritic cells hold complement-coated antigen in the germinal centre; B cells that bind it best are selected to become **plasma cells** and **memory B cells**. This is complement contributing directly to immunological memory.

6 — Cross-talk with the other plasma cascades

Complement is one of **four** plasma serine-protease cascades, and the other three are **coagulation**, **fibrinolysis** and the **kinin** system. They share substrates, share activators and — crucially — share inhibitors. Thrombin can cleave C3 and C5 directly; plasmin activates C3; factor XIIa activates the kinin system; and **C1-INH inhibits not only C1r/C1s and the MASPs but also kallikrein, factor XIIa and plasmin**.

WHY THIS APPARENTLY DRY FACT IS THE KEY TO THE NEXT SECTION

Hold two things together: **C1-INH restrains four cascades, not one**, and **bradykinin — the kinin system's product — is a powerful vasodilator that makes vessels leak**. Now predict what happens in a patient who has half the normal C1-INH. You should predict uncontrolled bradykinin, not uncontrolled complement. That prediction is correct, and it is the whole pathophysiology of hereditary angioedema. Do not memorise that disease — derive it.

6.4 Complement and Disease

Every complement disease in this lecture is one of three sentences: a component is missing, a regulator is missing, or activation is appropriate but excessive. Sort any clinical vignette into one of those three and the answer follows.

TABLE 6.5 COMPLEMENT DEFICIENCIES AND WHAT THEY PREDICT

What is missing	Clinical consequence	Why — derive it from the function
C1q, C1r, C1s, C4, C2 early classical components	Immune-complex disease — SLE-like illness, glomerulonephritis, vasculitis. C2 deficiency is the commonest complement deficiency in Europeans	Function 4 fails: no C3b on the complexes, no handle for erythrocyte CR1, so complexes circulate and precipitate in tissue. Infection risk is comparatively mild because the alternative pathway still delivers C3b onto microbes
C3	The most severe of all. Severe, recurrent pyogenic infection from infancy — <i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>S. aureus</i> , <i>Neisseria</i> — plus immune-complex disease	C3 is the single point through which all three pathways pass . Lose it and you lose opsonisation (function 2), the MAC (function 1), the anaphylatoxin C3a and the chemotaxin C5a (function 3), immune-complex clearance (function 4) and the C3d co-receptor signal (function 5). Five of the six functions fail at once — that is the whole explanation for its severity
C5, C6, C7, C8, C9 terminal components	Recurrent <i>Neisseria</i> infection — meningococcal meningitis and disseminated gonococcal infection — and remarkably little else	Only function 1 is lost. Opsonisation still works, so most bacteria are still handled — except <i>Neisseria</i> , which is a Gram-negative diplococcus that resists phagocytosis and against which the MAC is the main defence. Recurrent meningococcal disease = check the terminal components
MBL	Recurrent infection in infancy , often mild; frequently asymptomatic in adults	The lectin pathway matters most in the window between the loss of maternal IgG and the maturation of the infant's own antibody response

What is missing	Clinical consequence	Why — derive it from the function
Factor H or factor I	Atypical haemolytic uraemic syndrome, C3 glomerulopathy, and a secondary C3 deficiency with pyogenic infection	The alternative pathway never switches off, so C3 is consumed to exhaustion. The patient ends up phenotypically C3-deficient although the C3 gene is normal
DAF + CD59 acquired, PNH	Chronic intravascular haemolysis, haemoglobinuria, thrombosis	§6.2 — the host cell has lost its fireproofing and physiological tickover lyses it
C1-INH	Hereditary angioedema — recurrent, localised, non-itchy tissue swelling	Not a complement disease in its symptoms — see the card below

— ABSENCE PROVES FUNCTION — C3 DEFICIENCY, THE LECTURER'S OWN EMPHASIS

The slide says only this: “C3 deficiencies — with the most severe clinical manifestations.” You now have the reason, and the reason is the architecture of Part 5. **All three pathways exist to make one enzyme, and that enzyme exists to cleave C3.** Everything upstream is redundant — lose the classical pathway and the alternative still works. Nothing downstream of C3 is redundant, because there is nothing else that can produce C3b. C3 is the **single non-redundant node** of the entire system, and a deficiency there is functionally a deficiency of the whole of complement. This is why the organising idea of Part 5 was worth insisting on.

DISEASE CARD · C1-INH

Hereditary angioedema (HAE)

WHAT IT IS

An **autosomal dominant** deficiency of **C1 esterase inhibitor** — either too little protein (type I, ~85%) or a normal amount of a non-functional protein (type II). One working gene copy is not enough, because C1-INH works stoichiometrically and is consumed as it acts.

WHAT ACTUALLY CAUSES THE SWELLING

Not complement. **Bradykinin**.

Because: C1-INH is the main brake on the contact system as well as on C1. Without it, factor XIIa and kallikrein run unopposed, kallikrein cleaves high-molecular-weight kininogen, and **bradykinin** is released. Bradykinin acts on endothelial B2 receptors and makes post-capillary venules leak. C4 is consumed at the same time — which is why **a low C4 is the screening test** — but the low C4 is a marker of the defect, not the cause of the symptoms.

WHAT THE ATTACK LOOKS LIKE

Localised, non-pitting swelling of skin, lips, face, extremities, genitalia.

No urticaria, no itch — this is the single most useful discriminator at the bedside.

Abdominal attacks — bowel-wall oedema causing colic, vomiting and ascites, frequently mistaken for a surgical abdomen.

Laryngeal oedema — the cause of death, and the reason this is an emergency.

Attacks build over hours and last 2–5 days; triggers include trauma, dental work and stress. **ACE inhibitors are contraindicated** — ACE is the enzyme that degrades bradykinin.

REMEMBER IT AS

A complement gene causing a **kinin** disease. The screening test is complement (low C4); the symptoms, the drugs and the danger are all bradykinin.



Fig 6.13 Hereditary angioedema. Note what is **not** there: no weals, no erythematous urticarial plaques, no excoriation from scratching. The swelling is deep, of normal skin colour and sharply localised — the appearance of **oedema**, not of an allergic reaction.

EXAM TRAP — HEREDITARY VERSUS ALLERGIC ANGIOEDEMA

	Hereditary (C1-INH)	Allergic (IgE/histamine)
Mediator	Bradykinin	Histamine
Urticaria / itch	Absent	Present
Onset	Hours, lasting days	Minutes, resolving in hours
Family history	Usually positive, autosomal dominant	Not characteristic
C4 level	Low , even between attacks	Normal
Response to antihistamine, steroid, adrenaline	None — the classic clue	Good
Treatment	C1-INH concentrate; icatibant (bradykinin B2 antagonist); ecallantide (kallikrein inhibitor)	Adrenaline, antihistamine, corticosteroid

Complement and infection — the two-edged part

The lecture closes the disease section with viruses, and the point is that complement is not uniformly good for the host.

TABLE 6.6 COMPLEMENT AND VIRAL INFECTION

What complement does	For the host	Explanation
Forms larger viral aggregates	Good	Many virions clumped into one particle means fewer infectious units , even though the total amount of virus is unchanged
Coats the particle with antibody and/or complement and blocks attachment	Good	The receptor-binding site is physically covered, so the virion cannot dock on a susceptible cell
Lyses most enveloped viruses	Good	The envelope is a lipid bilayer, so the MAC works on it exactly as it works on a Gram-negative outer membrane
Facilitates binding of the coated particle to cells bearing Fc or CR1	Bad	The opsonin becomes a door key . A virus that can replicate in macrophages is handed a new route in — the basis of antibody-dependent enhancement , best known in dengue
Drives inflammation at the site	Both	Necessary for clearance, but excessive activation is itself the injury in ischaemia-reperfusion, ARDS and sepsis. This is why anti-C5 therapy exists

THE SUMMARY THE LECTURER ASKS FOR

The final slide of this lecture sets two tasks: “Review the differences between the three activation pathways of complement” and “Function of complement and its regulation.” Both are answered in full by the master tables in Part 7 — go there next and use them as the revision sheet.

PART SEVEN

07

Master Tables, Self-Test and Glossary

This Part contains no new material. It is the compression: everything in Parts 1–6 reduced to two master tables, a set of deliberate look-alikes, twenty-four questions and a glossary. On the night before the exam, read nothing else.

HOW TO USE THIS PART

- Cover the right-hand columns of Table 7.1 and 7.2 and rebuild them from the left-hand column alone.
- Do the twenty-four questions **before** re-reading anything; the ones you miss tell you which section to return to.
- Every answer carries a section reference. Follow it — that is what it is for.

7.1 Master Table — The Five Immunoglobulins

TABLE 7.1 THE FIVE CLASSES, SIDE BY SIDE

	IgG	IgM	IgA	IgE	IgD
Heavy chain	γ (gamma)	μ (mu)	α (alpha)	ε (epsilon)	δ (delta)
Form in serum	Monomer	Pentamer (+ J chain)	Monomer; di-mer in secretions (+ J chain + SP)	Monomer	Monomer
C _H domains	3 + hinge	4 , no hinge	3 + hinge	4 , no hinge	3 + long hinge
Serum level	Highest (~75–80% of total Ig)	~10%	~15% in serum; most abundant overall in the body once secretions are counted	Lowest	Trace
Complement	Yes — classical, via C _H 2 (IgG1–3)	Yes — the best , via C _H 3; one molecule is enough	Not classical; lectin / alternative	No	No
Crosses placenta	Yes — the only one (FcRn)	No	No	No	No
Defining job	The workhorse of the secondary response: opsonisation, neutralisation, ADCC, neonatal immunity	The first responder : primary response, BCR on naïve B cells, complement fixation	The gatekeeper : mucosal surfaces, milk, tears, saliva, gut	The alarm : mast-cell arming, type I hypersensitivity, anti-parasitic	The co-receptor : with IgM on the naïve B-cell surface
Absence proves it	Low IgG at 3–6 months (transient hypogammaglobulinemia of infancy)	IgM in cord blood = intrauterine infection	Selective IgA deficiency: sinopulmonary and GI infection; anti-IgA anaphylaxis risk	High IgE: atopy, hyper-IgE syndrome	No recognised deficiency syndrome
Remember it as	G for G estation and the G eneral-purpose antibody	M for M acro and iM mediate	A for A rea guarded — every wet surface	E for E osinophil and E xaggerated response	D for D on't know much — and that is honest

7.2 Master Table — The Three Pathways Compared

THIS IS THE LECTURER'S STATED EXAM OBJECTIVE

The final slide asks, in these words, for “*the differences between the three activation pathways of complement.*” Table 7.2 is that answer. Learn it row by row, not pathway by pathway — the rows are the questions an examiner can ask.

TABLE 7.2 CLASSICAL VS LECTIN VS ALTERNATIVE

	Classical CP	Lectin (MBL) LP	Alternative AP
Activator	Antigen-antibody complex — IgM (C _H 3) or IgG1-3 (C _H 2). Bound antibody only	Microbial carbohydrate — mannose, fucose, GlcNAc on the pathogen surface	The pathogen surface itself — LPS (endotoxin), polysaccharides, zymosan, teichoic acid
Recognition molecule	C1q (in the C1 complex, C1qr ₂ s ₂)	MBL or ficolins — structurally like C1q	None — there is no recognition step. C3 hydrolyses spontaneously (tickover) and the surface merely decides whether the product survives
Initiating enzyme	C1r → C1s	MASP-1 and MASP-2	Factor D (cleaves factor B)
Components used	C1q, C1r, C1s, C4, C2 , C3	MBL, MASP, C4, C2 , C3	Factors B, D, properdin , C3 — letters, not C4 and C2
C3 convertase	C4b2a	C4b2a — identical to classical	C3bBb (stabilised by properdin)
C5 convertase	C4b2a3b	C4b2a3b	C3bBb3b
Antibody required	Yes	No	No
Immunity	Adaptive — needs a prior response	Innate	Innate
Timing in an infection	Late — days, once antibody exists	Early — hours to days (MBL is an acute-phase protein)	Immediate — minutes, running before the pathogen even arrives
Evolutionary age	Youngest , despite the name “classical”	Intermediate	Oldest
Distinctive feature	Needs two IgG or one IgM; soluble antibody cannot do it	C1q's job done by a lectin — pattern recognition instead of antibody	Amplification loop : C3b makes more C3bBb which makes more C3b. Also amplifies the other two pathways
Key regulator	C1-INH , C4bp, DAF, MCP, CR1	C1-INH (on the MASPs), C4bp, DAF, MCP	Factor H + factor I, DAF, CR1
What is the same in all three	Everything from C3 onwards. All three build a C3 convertase, all three cleave C3 to C3a + C3b, all three end in the same terminal pathway: C5 → C5b + C5a, then C6, C7, C8, C9 → the MAC.		

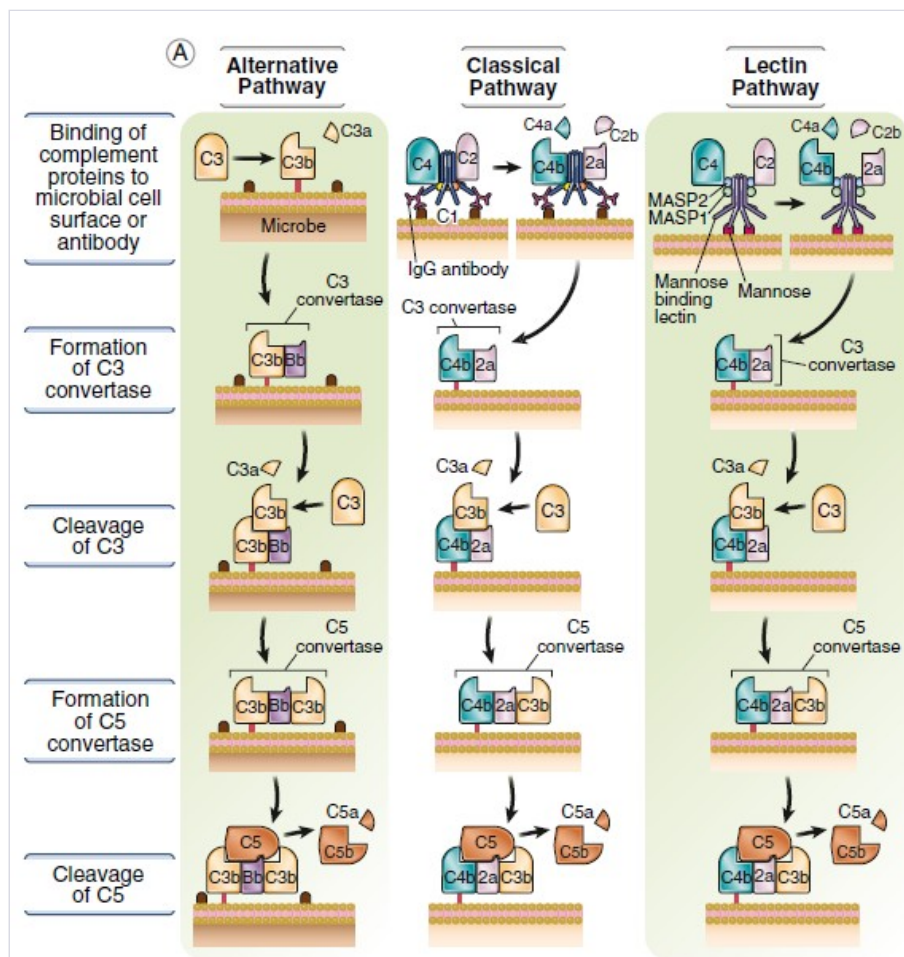


Fig 7.1 Table 7.2 drawn. The three pathways run down the page in parallel columns and the four stages run across in rows — binding, formation of the C3 convertase, cleavage of C3, formation of the C5 convertase, cleavage of C5. Read it **across** a row and you get the differences; read it **down** a column and you get one pathway end to end. Notice that the bottom two rows are the same picture three times: that is the convergence.

THE ONE-LINE VERSION OF THE WHOLE COMPLEMENT SYSTEM

Three doors, one room. Antibody opens one door, sugar opens another, and the third has no door at all — it is always ajar. Every door leads to the same room, and the room is **C3**.

7.3 Tell Them Apart — the Look-Alikes

These are the pairs that get confused, and the exam knows it. In each row the question is the same: **what single fact separates them?**

TABLE 7.3 TWELVE CONFUSABLE PAIRS AND THE FACT THAT SEPARATES THEM

The pair	The separating fact	Where
Antibody vs immunoglobulin	Ig is the structure ; antibody is the function . All antibodies are Ig; a myeloma paraprotein is Ig that is not an antibody	§1.1
Papain vs pepsin	Papain cuts above the disulphides → 2 Fab + Fc (monovalent pieces). Pepsin cuts below → F(ab') ₂ + pFc' (still bivalent)	§1.7

The pair	The separating fact	Where
Isotype vs allotype vs idio-type	Species (all humans share it) vs individual (allelic, inherited) vs clone (the V region itself). Isotype and allotype are in C ; idio-type is in V	§2.3
CDR vs framework	CDRs are the loops that touch antigen (3 per chain, 6 per binding site); FRs are the β-sheet scaffold that holds them in place	§1.4
J chain vs secretory piece	Both are extra polypeptides on polymeric Ig, but the J chain is made by the plasma cell and the secretory piece is made by the epithelial cell and added during transcytosis	§1.6
C2a vs C2b	C2 is the only exception to “a = small”. In this course the large fragment is C2a and the convertase is C4b2a	§5.1
C3b vs iC3b	Both are opsonins. C3b builds convertases and binds CR1 ; iC3b cannot build anything and binds CR3/CR4. Factor I + a cofactor makes the change, and it is irreversible	§6.2, §6.3
DAF vs MCP vs CD59	Decay the convertase vs cofactor for factor I vs block C9 . First two act at C3; the third acts at the MAC	§6.2
C3a/C4a vs C5a	All three are anaphylatoxins ; only C5a is also a chemotaxin , and it is the most potent of the three	§6.3
CR1 vs CR2	CR1/CD35 binds C3b — regulation, opsonisation, erythrocyte transport. CR2/CD21 binds C3d — the B-cell co-receptor and FDC antigen retention	§6.3
Complement low vs complement high	Low = being consumed (active immune-complex disease, SLE). High = acute-phase response. Same molecule, opposite meanings	§5.1, §6.3
Hereditary vs allergic angioedema	Bradykinin, no itch, low C4, no response to adrenaline vs histamine, urticarial, normal C4, responds	§6.4

7.4 Self-Test — 24 Questions with Answers

Answer each one out loud or on paper before reading on. An answer you can recognise but not produce is not learned.

Q1 A patient's serum contains a large amount of a single immunoglobulin with no detectable binding to any antigen. Is it an antibody?

No. It is an immunoglobulin but not an antibody. *Immunoglobulin* names a structure; *antibody* names a function — an Ig that binds a specific antigen. A monoclonal paraprotein in myeloma is the standard example. (§1.1)

Q2 How many CDRs are there in one antigen-binding site, and where do they come from?

Six — three from the V_H and three from the V_L . The heavy and light variable domains pair, and their six hypervariable loops come together at one end to form the surface that contacts the epitope. (§1.4)

Q3 Antivenom is manufactured as $F(ab')_2$ rather than whole IgG. Give the two reasons.

$F(ab')_2$ keeps **both** binding sites, so it is still bivalent and can cross-link and neutralise toxin — unlike a single Fab. And it has **no Fc**, so it cannot fix complement or engage Fc receptors, which removes most of the serum-sickness and anaphylaxis risk of injecting a foreign whole antibody. Pepsin gives exactly this fragment. (§1.7)

Q4 The genome has about 2×10^4 genes but you can make about 10^{11} different antibodies. Name the four mechanisms that bridge the gap.

(1) **Combinatorial V(D)J recombination** — choosing one of many V, D and J segments; (2) **junctional diversity** — imprecise joining plus N and P nucleotide addition, which is why CDR3 is the most variable loop; (3) **combinatorial heavy-light pairing** — any H can pair with any L, so the two repertoires multiply; (4) **somatic hypermutation** during the germinal-centre response. Tonegawa won the 1987 Nobel Prize for the first. (§2.1, §2.2)

Q5 Two IgG molecules from the same person differ in a constant-region amino acid inherited from different parental alleles. What is that difference called?

Allotype. Isotype would be a difference between classes (present in all humans); idiotypic would be a difference in the V region between clones. Allotype is the **allelic, inherited** variation within a species, found in the C region. (§2.3)

Q6 Which immunoglobulin crosses the placenta, and what does a newborn's antibody profile look like at 3 months?

IgG only, actively transported by **FcRn**. At 3 months the infant has a falling level of maternal IgG and a still-immature output of its own, which produces the nadir known as **transient hypogammaglobulinaemia of infancy** — the reason this age is a window of vulnerability. (§3.2)

Q7 IgM is found in the cord blood of a newborn. What does this mean and why?

Intrauterine infection. IgM is a pentamer and does not cross the placenta — there is no transport receptor for it — so any IgM in the newborn must have been synthesised by the fetus. Detecting specific IgM in cord blood is therefore diagnostic of congenital infection. (§3.2)

Q8 A patient with selective IgA deficiency develops anaphylaxis during a blood transfusion. Explain.

Having never made IgA, the patient's immune system treats it as **foreign** and some patients develop **anti-IgA antibodies**. Blood products contain donor IgA, so the transfusion delivers an antigen against which the patient is already sensitised, producing an anaphylactic reaction. Such patients need **washed** red cells or IgA-deficient donor products. (§3.2)

Q9 Read the monoclonal antibody names infliximab, trastuzumab and adalimumab. What does each suffix tell you, and why does it matter clinically?

-ximab = chimeric (mouse V on human C), **-zumab** = humanised (only the CDRs are murine), **-umab** = fully human. The gradient is how much mouse protein remains, and that predicts **immunogenicity** — the more murine sequence, the more likely the patient makes anti-drug antibodies that cause infusion reactions and loss of response. (§4.1)

Q10 Why can a hybridoma grow in HAT medium when neither parent cell can?

The myeloma partner is **HGPRT-deficient**, so it cannot use the salvage pathway that HAT forces cells onto, and it dies. The spleen B cell has HGPRT but is **mortal** and dies in culture within days. Only the fusion product has **both** properties — HGPRT from the B cell and immortality from the myeloma — so only hybridomas survive. Milstein and Köhler, Nobel Prize 1984. (§4.1)

Q11 Decode the symbol C4b2a3b without having memorised it.

Read it as an assembly list: C4b (large fragment of C4, surface-bound) + 2a (large fragment of C2) + 3b (a C3b). C4b2a alone is the **classical C3 convertase**; adding a C3b shifts the substrate specificity, so C4b2a3b is the **classical/lectin C5 convertase**. The same logic applied to letters gives C3bBb (AP C3 convertase) and C3bBb3b (AP C5 convertase). (§5.1)

Q12 Why can soluble IgG not activate complement, while the same IgG bound to a cell surface can?

C1q must engage **at least two of its six globular heads at the same time** to change conformation and activate C1r and C1s. In solution IgG molecules are far apart and randomly oriented, so two Fc regions are never close enough. On a surface the antibodies are fixed, dense and aligned, and C1q can bridge two adjacent C_{H2} regions. Complement activation is therefore **geometrically restricted to genuine targets** — no separate safeguard is needed. (§5.3)

Q13 One IgM molecule can fix complement but two IgG molecules are needed. Why?

Because of the same two-head rule. A pentameric IgM already carries **five Fc regions clustered in one molecule**, so a single bound IgM presents C1q with adjacent binding sites once it has flipped from the planar “star” to the bent “staple” conformation. An IgG monomer has only one Fc, so two separate molecules must land close together by chance. This is why IgM is the most efficient complement-fixing isotype. (§3.2, §5.3)

Q14 Give the activator, the recognition molecule and the C3 convertase of each of the three pathways.

Classical: antigen–antibody complex → C1q → C4b2a. **Lectin:** microbial mannose/fucose/GlcNAc → MBL or ficolin with MASPs → C4b2a. **Alternative:** the pathogen surface itself (LPS, polysaccharide) → no recognition molecule, spontaneous C3 tickover → C3bBb, stabilised by properdin. (§5.3–§5.5, Table 7.2)

Q15 The alternative pathway is constantly active in your plasma. Why does it not destroy your own cells?

Because your cells are **protected**, not because complement avoids them. Host membranes carry **sialic acid and heparan sulphate**, which recruit **factor H**; factor H decays C3bBb and hands C3b to **factor I**. They also carry **DAF** (decay) and **MCP** (cofactor). Microbial surfaces have none of this, so on them the C3b survives and the amplification loop runs. Self/non-self discrimination in the alternative pathway is **protection of self**, not recognition of non-self. (§5.5, §6.1)

Q16 Describe how the MAC kills a cell, in five steps.

(1) A C5 convertase cleaves **C5 → C5a + C5b**. (2) **C6** and **C7** bind C5b; C7 exposes a hydrophobic site and the complex can now enter the **lipid bilayer**. (3) **C8** binds and pushes into the membrane, forming a small leaky pore. (4) **C9** binds C8 and **polymerises** — 10 to 16 molecules — into a ring. (5) The ring is a **transmembrane channel**: ions and water flow in, the cell swells and bursts by **osmotic lysis**. (§5.6)

Q17 Name the three points in the cascade at which regulators act, with one regulator for each.

(1) **Recognition** — C1-INH inhibits C1r₂s₂ and the MASPs. (2) **The C3/C5 convertase** — DAF decays it; MCP, CR1, C4bp and factor H act as cofactors for factor I. (3) **The MAC** — S protein and clusterin cap C5b67 in the fluid phase; CD59 blocks C9 polymerisation on the membrane. (§6.1, Table 6.1)

Q18 A cell carries both DAF and MCP. Since both stop the same convertase, why does it need both?

They do different things. **DAF decays**: it knocks C2a or Bb off, stopping the enzyme now but leaving an intact C4b or C3b that can recruit a fresh partner and restart. **MCP is a cofactor**: it presents that C4b/C3b to **factor I**, which cleaves it into fragments that can never form a convertase again. DAF is reversible and immediate; MCP is slower and permanent. (§6.2)

Q19 A patient has chronic intravascular haemolysis with haemoglobinuria. Complement levels, all three pathways and C3 are normal. What is the defect and what is the treatment?

Paroxysmal nocturnal haemoglobinuria. An acquired **PIGA** mutation in a haematopoietic stem cell abolishes the **GPI anchor**, so the clone lacks every GPI-linked membrane protein — including **DAF (CD55)** and **CD59**. Nothing is wrong with complement; the red cells have simply lost their protection, and the normal physiological tickover of the alternative pathway lyses them. Treatment is **eculizumab**, an anti-C5 monoclonal, which blocks the terminal pathway. (§6.2)

Q20 Name the six functions of complement, and the molecule responsible for each.

(1) **Cytolysis** — C5b–C9 (MAC). (2) **Opsonisation** — C3b, C4b, iC3b, read by CR1, CR3, CR4. (3) **Inflammation** — the anaphylatoxins C5a, C3a, C4a; C5a is also the chemotaxin. (4) **Immune-complex clearance** — C3b gripped by CR1 on erythrocytes. (5) **Mediating adaptive immunity** — C3d binding CR2/CD21 on B cells and FDCs. (6) **Cross-talk** with the coagulation, fibrinolytic and kinin systems. (§6.3, Table 6.3)

Q21 Complement causes immune-complex disease, and complement deficiency also causes immune-complex disease. Resolve the contradiction.

There is no contradiction, because complement has **two roles** with respect to immune complexes. Normally C3b deposited on a complex breaks up the lattice and gives erythrocyte **CR1** a handle to carry the complex to liver and spleen — so complement **clears** complexes. If the early classical components are deficient, no C3b is deposited, nothing is cleared, and complexes precipitate in glomeruli and vessel walls. Once deposited, complement activation at the site is what **causes the tissue injury**. Hence deficiency of C1q, C1r, C1s, C4 or C2 produces a lupus-like disease. (§6.3, §6.4)

Q22 Why is C3 deficiency the most severe complement deficiency, and why does terminal-component deficiency present specifically with *Neisseria*?

C3 is the single non-redundant node. All three pathways exist only to build an enzyme that cleaves C3, and nothing downstream can be reached without it — so losing C3 loses opsonisation, the MAC, the anaphylatoxins, immune-complex clearance and the C3d co-receptor signal simultaneously. **Terminal deficiency loses only lysis.** Opsonisation still works, and opsonisation handles most bacteria — but not *Neisseria*, a Gram-negative diplococcus that resists phagocytosis and is killed mainly by the MAC. **Recurrent meningococcal disease should always prompt measurement of the terminal components.** (§6.4)

Q23 A young woman has recurrent episodes of non-itchy facial and laryngeal swelling and unexplained abdominal colic. Antihistamines and adrenaline do nothing. Her father had the same. What is the diagnosis, the screening test, and the mediator?

Hereditary angioedema, autosomal dominant deficiency of **C1-INH**. The screening test is a **low C4** — persistently low, even between attacks — with confirmation by C1-INH level and function. The mediator is **bradykinin**, not histamine: C1-INH also restrains factor XIIa and kallikrein, so its loss releases bradykinin, which makes venules leak. That is why antihistamines, steroids and adrenaline fail and why ACE inhibitors are contraindicated. Treat with **C1-INH concentrate** or **icatibant**. (§6.4)

Q24 Give one way in which complement helps a virus infect the host.

By coating the virion with antibody and/or complement, complement **facilitates its binding to cells bearing Fc receptors or CR1**. For a virus that can replicate inside macrophages, the opsonin becomes an entry route — the opsonin is a door key rather than a death sentence. This is the basis of **antibody-dependent enhancement**, best known in dengue. All the other listed effects — aggregation, blocked attachment, lysis of enveloped viruses — favour the host. (§6.4, Table 6.6)

7.5 Glossary of Key Terms

Allotype

Inherited allelic variation in the constant region; differs between individuals of one species.

Anaphylatoxin

A small complement fragment (C5a, C3a, C4a) that triggers mast-cell degranulation and inflammation. Potency C5a > C3a > C4a.

Antibody

An immunoglobulin that binds a specific antigen. The functional term.

C1-INH

Serpin inhibiting C1r/C1s, the MASPs, factor XIIa, kallikrein and plasmin. Deficiency → hereditary angioedema.

C3 convertase

The enzyme that cleaves C3. C4b2a (classical, lectin) or C3bBb (alternative). The point of the whole system.

CDR

Complementarity-determining region; a hypervariable loop that contacts antigen. Three per chain, six per binding site.

Chemotaxin

A molecule that directs cell migration along its concentration gradient. In complement this means **C5a**.

Complement

A heat-labile system of serum glycoproteins that opsonise pathogens and induce inflammation. Destroyed at 56 °C for 30 min.

CR1 / CR2 / CR3 / CR4

Complement receptors. CR1 binds C3b; CR2 binds C3d; CR3 and CR4 bind iC3b.

DAF (CD55)

GPI-anchored membrane protein that accelerates decay of C3 convertases on host cells.

Factor H

Soluble regulator that binds sialic acid on host surfaces, decays C3bBb and acts as cofactor for factor I. The main reason the alternative pathway tolerates self.

Factor I

Serine protease that cleaves C3b to iC3b and C4b to C4c + C4d. Requires a cofactor. The only irreversible regulatory step.

Fab / F(ab')₂ / Fc

Papain gives 2 Fab + Fc; pepsin gives F(ab')₂ + pFc'. Fab binds; Fc acts.

Hinge region

Proline-rich flexible segment between C_H1 and C_H2 allowing the arms to open from 0° to 90°. Absent in IgM and IgE.

Hybridoma

Fusion of an antibody-producing B cell with an immortal myeloma cell; the source of monoclonal antibodies.

iC3b

C3b cleaved by factor I. Still an opsonin (via CR3/CR4) but can no longer build a convertase.

Idiotypic

The set of antigenic determinants in the V region, unique to one clone.

Isotype

The antibody class defined by the heavy-chain constant region: γ , μ , α , ϵ , δ . Shared by all members of a species.

J chain

Small polypeptide made by the **plasma cell** that joins monomers into pentameric IgM and dimeric IgA.

MAC

Membrane attack complex, C5b–C9. A transmembrane channel causing osmotic lysis.

MASP

MBL-associated serine protease; the lectin pathway's equivalent of C1r and C1s.

MBL

Mannan-binding lectin; an acute-phase protein made by hepatocytes, structurally like C1q, that binds mannose, fucose and GlcNAc on microbes.

MCP (CD46)

Membrane cofactor protein; presents C3b/C4b on host cells to factor I.

CD59 (MIRL)

GPI-anchored membrane protein that blocks C9 polymerisation and so prevents MAC formation on host cells.

Opsonin

A molecule that coats a particle and makes it easier to phagocytose. C3b, C4b, iC3b, IgG.

PNH

Paroxysmal nocturnal haemoglobinuria; acquired PIGA mutation → loss of GPI anchor → loss of CD55 and CD59 → complement-mediated intravascular haemolysis.

Properdin (factor P)

The only **positive** regulator of complement; stabilises C3bBb on microbial surfaces.

Secretory piece

Fragment of the poly-Ig receptor, made by the **epithelial cell**, that carries dimeric IgA across the mucosa and protects it from proteases.

Serpin

Serine protease inhibitor acting as a suicide substrate. C1-INH is the complement example.

Somatic hypermutation

Point mutation of rearranged V genes in the germinal centre, producing affinity maturation.

Tickover

Continuous spontaneous hydrolysis of C3 in plasma, which keeps the alternative pathway permanently primed.

V(D)J recombination

Somatic rearrangement of V, D and J gene segments that creates the variable region. Tonegawa, Nobel Prize 1987.

LAST WORD

Two sentences carry this whole lecture. **An antibody is one molecule with two ends** — a variable end that decides *what*, and a constant end that decides *what happens next*. **Complement is thirty proteins with one event** — the cleavage of C3 — reached by three doors and restrained at three points. Everything else in these seven Parts is detail hung on those two frames.